Caspian Journal of Applied Mathematics, Ecology and Economics
V. 11, No 2, 2023, December
ISSN 1560-4055

Inverse Boundary Value Problem for Benney-Luke Linearized Equation with Nonlocal Time-integral Conditions of Second Kind

B.K. Veliyeva

Abstract. In the paper we study an inverse boundary value problem with a time-dependent unknown coefficient for a linearized Benney-Luke equation with nonlocal time- integral conditions of second kind. The essence of the problem is to determine the unknown coefficient together with the solution. The problem is considered in a rectangular domain. When solving the original inverse problem, a transition is made from the original inverse problem to some auxiliary inverse problem. The existence and uniqueness of the solution of the auxiliary problem is proved by means of compressed mappings. Then the transition to the original inverse problem is made again, as a result a conclusion is made about the solvability of the original inverse problem.

Key Words and Phrases: inverse problem, Benney-Luke equation, existence and uniqueness of the classic solution.

2010 Mathematics Subject Classifications: 35R30

---

1. Introduction

Many problems of mathematical physics, mechanics of continuum are boundary value problems reduced to the integration of a differential equation or a system of partial equations under given boundary and initial conditions. Many problems of gas dynamics, theory of elasticity, theory of plates and shells are reduced to the consideration of higher order partial differential equations [1]. Differential equations of fourth order are of great interest from the point of view of applications (see, e.i. [2, 3]). Benney-Luke type partial differential equations have applications in mathematical physics (see [3]).

The problems in which together with the solution of one or another differential equation, it is necessary to determine the coefficient (coefficients) of the equation itself, or the right hand side of the equation, in mathematics and in mathematical modeling are called inverse problems. Theory of inverse problems for differential equations is a dynamically developing section of modern science. Last time, the inverse problems arise in very different fields of human activity as seismology, mineral exploration, biology, medicine, quality control of industrial products, etc. that places them among the pressing problems of modern mathematics. Different inverse problems for certain types of partial differential equations have been studied in many works. Here first of all we note the works of A.N.Tikhonov [4], M.M.Lavrent'ev [5, 6], V.K.Ivanov [7] and their followers. For more information you can read the monograph of A.M.Denisov [9].

Theory of inverse boundary value problems for fourth order equations still remain small studied. The works [**?** 10, 11, 12] have been devoted to the inverse boundary value problems for the Benney–Luke equation.

The goal of the present paper is to prove the existence and uniqueness of the solution of the inverse boundary value problem for the Benney-Luke equation with time non-local integral conditions of second kind.

2. Statement of the problem and its reduction to the equivalent problem

Let us consider for the equation

$$u_{tt}(x,t) - u_{xx}(x,t) + \alpha u_{xxxx}(x,t) - \beta u_{xxtt}(x,t) = a(t)u(x,t) + f(x,t), (x,t) \in D_T \quad (1)$$

in the domain $D_T = \{(x,t) : 0 \le x \le 1,\ 0 \le t \le T\}$, the inverse boundary value problem with the nonlocal initial conditions

$$u(x,0) = \int_0^T p_1(t)u(x,t)dt + \varphi(x), u_t(x,0) = \int_0^T p_2(t)u(x,t)dt + \psi(x) (0 \le x \le 1), \quad (2)$$

the boundary conditions

$$u_x(0,t) = 0, u(1,t) = 0, u_{xxxx}(0,t) = 0, u_{xx}(1,t) = 0 (0 \le t \le T) \quad (3)$$

and with the additional condition

$$u(0,t) = h(t)\,(0 \le t \le T), \quad (4)$$

where $\alpha > 0$, $\beta > 0$ are fixed numbers, $f(x,t)$, $p_1(t)$, $p_2(t)$, $\varphi(x)$, $\psi(x)$, $h(t)$ are the given functions $u(x,t)$, $a(t)$ are the desired functions

We introduce the denotation

$$\tilde{C}^{4,2}(D_T) = \{u(x,t) : u(x,t) \in C^2(D_T)\,, u_{ttx}(x,t),$$
$$u_{ttxx}(x,t), u_{xxx}(x,t), u_{xxxx}(x,t) \in C(D_T)\}\,.$$

Definition 1. *Under the classical solution of the inverse boundary value problem* (1)-(4) *we understand the pair* $\{u(x,t)\,,\ a(t)\}$*of functions* $u(x,t) \in \tilde{C}^{4,2}(D_T)$, $a(t) \in C[0,T]$ *satisfying the equation* (1) *and conditions* (2)-(4) *in the usual sense.*

To study the problem (1)-(4) at first we consider the following problem:

$$y''(t) = a(t)\,y(t) \quad (0 \le t \le T), \quad (5)$$

$$y(0) = \int_0^T p_1(t)y(t)dt, \; y'(0) = \int_0^T p_2(t)y(t)dt, \tag{6}$$

where $a(t) \in C[0,T]$, $p_1(t)$, $p_2(t) \in C[0,T]$ are the given functions, $y = y(t)$ is an unknown function and under the solution of problem (5), (6) we understand the function $y(t)$ belonging to $C^2[0,T]$ and satisfying condition of (5), (6) in the usual sense.

The following lemma is valid

Lemma 1 ([13]). *Let us functions $p_1(t) \in C[0,T], p_2(t) \in C[0,T]$, $a(t) \in C[0,T]$ and*

$$\|a(t)\|_{C[0,T]} \leq R = const.$$

Furthermore

$$\left(T\,\|p_2(t)\|_{\,C[0,T]} + \|p_1(t)\|_{\,C[0,T]} + \frac{T}{2}R\right)T < 1.$$

Then problem (5), (6) *has only a trivial solution.*

Along with the inverse boundary value problem (1)- (4) we consider the following auxiliary inverse boundary value problem:

It is required to determine the pair $\{u(x,t), a(t)\}$ of functions $u(x,t) \in \tilde{C}^{4,2}(D_T)$, $a(t) \in C[0,T]$ from the relations (1)-(4) ,

$$h''(t) - u_{xx}(0,t) + \alpha u_{xxxx}(0,t) - \beta u_{xxtt}(0,t) = a(t)h(t) + f(0,t) \; (0 \leq t \leq T) \tag{7}$$

The following theorem is valid

Theorem 1. *Let $\varphi(x)$, $\psi(x) \in C[0,1]$, $p_i(t) \in C[0,T]$ $(i = 1,2)$, $f(x,t) \in C(D_T)$, $h(t) \in C^2[0,T]$, $h(t) \neq 0$ $(0 \leq t \leq T)$ and the following agreement conditions be fulfilled:*

$$\varphi(0) = h(0) - \int_0^T p_1(t)h(t)dt, \quad \psi(0) = h''(0) - \int_0^T p_2(t)h(t)dt.$$

Then the following statements are valid:

A. *Each classic solution $\{u(x,t), a(t)\}$ of problem* (1)-(4) *is the solution of problem* (1)-(3), (7) *as well;*

B. *Each solution of $\{u(x,t), a(t)\}$ of problem* (1)-(3), (7), *such that*

$$\left(T\,\|p_2(t)\|_{\,C[0,T]} + \|p_1(t)\|_{\,C[0,T]} + \frac{T}{2}\,\|a(t)\|_{C[0,T]}\right)T < 1 \tag{8}$$

is the classic solution (1)-(4).

Proof. Let $\{u(x,t), a(t)\}$ be a classic solution of problem (1)-(4). Substituting $x = 0$ in equation (1), we find:

$$u_{tt}(0,t) - u_{xx}(0,t) + \alpha u_{xxxx}(0,t) - \beta u_{xxtt}(1,t) =$$

$$= a(t)u(0,t) + f(0,t) \ \ (0 \le t \le T)\,, \tag{9}$$

In what follows, assuming $h(t) \in C^2[0,T]$ and differentiating twice (4), we have:

$$u_t(0,t) = h'(t)\,,\ u_{tt}(0,t) = h''(t)\,(0 \le t \le T).$$

Taking into account these relations, from (9) allowing for (4) we obtain the fulfillment of (7). Now, we assume that $\{u(x,t), a(t)\}$ is the solution of problem (1)- (3), (7). Then from (7) and (9) we obtain:

$$\frac{d^2}{dt^2}(u(0,t) - h(t)) = a(t)(u(0,t) - h(t)) \ \ (0 \le t \le T). \tag{10}$$

By virtue of (2) and agreement conditions $\varphi(0) = h(0) \ - \int_0^T p_1(t)h(t)dt$,
$\psi(0) = h''(0) - \int_0^T p_2(t)h(t)dt$ we have:

$$u(0,0) - h(0) - \int_0^T p_1(t)(u(0,t) - h(t))dt = u(0,0) - \int_0^T p_1(t)u(0,t)dt -$$

$$= \varphi(0) - \left(h(0) - \int_0^T p_1(t)h(t)dt\right) = 0,$$

$$u_t(0,0) - h'(0) - \int_0^T p_2(t)(u(0,t) - h(t))dt = u_t(0,0) - \int_0^T p_2(t)u(0,t)dt -$$

$$= \varphi(0) - \left(h'(0) - \int_0^T p_2(t)h(t)dt\right) = 0. \tag{11}$$

From (10), (11) by lemma 1 we conclude that condition (4) is fulfilled. The theorem is proved.

3. Studying the existence and uniqueness of the classic solution of the inverse boundary value problem

We will look for the first component $u(x,t)$ of the solution $\{u(x,t), a(t)\}$ of problem (1)-(3), (7) in the form:

$$u\,(x,t) = \sum_{k=1}^{\infty} u_k\,(t) \cos \lambda_k x \ \ (\lambda_k = \frac{\pi}{2}(2k = 1)), \tag{12}$$

where

$$u_k(t) = 2\int_0^1 u(x,t) \cos \lambda_k x \, dx.$$

Applying the formal scheme of the Fourier method, from (1), (2) we obtain:

$$(1 + \alpha\lambda_k^2)u_k''(t) + \lambda_k^2(1 + \alpha\lambda_k^2)u_k(t) = F_k(t; u, a, b) \ \ (0 \le t \le T;\ k = 1, 2, ...), \tag{13}$$

$$u_k(0)=\int_0^T p_1(t)u_k(t)dt+\varphi_k,\ \ u'_k(0)=\int_0^T p_2(t)u_k(t)dt+\psi_k\ \ (k=1,2,...), \tag{14}$$

where

$$F_k(t;u,a)=f_k(t)+a(t)u_k(t), f_k(t)=2\int_0^1 f(x,t)\cos\lambda_k x dx,$$

$$\varphi_k=2\int_0^1\varphi(x)\cos\lambda_k x dx, \psi_k=2\int_0^1\psi(x)\cos\ \lambda_k x dx\ \ (k=1,2,...).$$

Solving the problem (13), (14), we find:

$$u_k(t)=\left(\int_0^T p_1(t)u_k(t)dt+\varphi_k\right)\cos\beta_k t+\frac{1}{\beta_k}\left(\int_0^T p_2(t)u_k(t)dt+\psi_k\right)\sin\beta_k t+$$

$$+\frac{1}{\beta_k(1+\beta\lambda_k^2)}\int_0^t F_k(\tau\,;u,a)\ \sin\beta_k\,(t-\tau)\ d\tau\,. \tag{15}$$

where

$$\beta_k=\lambda_k\sqrt{\frac{1+\alpha\lambda_k^2}{1+\beta\lambda_k^2}}(k=1,2,...).$$

After substituting the expression $u_k(t)$ $(k=1,2,...)$ from (15) to (12), for determining the component $u\,(x,t)$ of the solution of problem (1)-(3), (7) we obtain:

$$u(x,t)=\sum_{k=1}^{\infty}\left\{\left(\int_0^T p_1(t)u_k(t)dt+\varphi_k\right)\cos\beta_k t+\frac{1}{\beta_k}\left(\int_0^T p_2(t)u_k(t)dt+\psi_k\right)\sin\beta_k t+\right.$$

$$\left.+\ \frac{1}{\beta_k(1+\beta\lambda_k^2)}\int_0^t F_{2k-1}(\tau\,;u,a,b)\ \sin\beta_k\,(t-\tau)\ d\tau\right\}\cos\ \lambda_k x. \tag{16}$$

Now, from (7) allowing for (15), we obtain:

$$a(t)h(t)\ =h''(t)\ -f(0,t)+$$

$$+\sum_{k=1}^{\infty}(\lambda_k^2(1+\alpha\lambda_k^2)u_k(t)+\beta\lambda_k^2u''_k(t)).$$

or taking into account

$$u''_k(t)=-\frac{\lambda_k^2(1+\alpha\lambda_k^2)}{1+\beta\lambda_k^2}u_k(t)+\frac{1}{1+\beta\lambda_k^2}F_k\,(t;u,a)$$

we have:

$$a(t)h(t)\ =h''(t)\ -f(0,t)+$$

$$+\sum_{k=1}^{\infty}\left(\beta_k^2u_k(t)+\frac{\beta\lambda_k^2}{1+\beta\lambda_k^2}F_k\,(t;u,a)\right)\,. \tag{17}$$

Now from (17) we find:

$$a(t) = [h(t)]^{-1}\left\{h''(t) \;-f(0,t)+\right.$$

$$\left.++\sum_{k=1}^{\infty}\left(\beta_k^2 u_k(t)+\frac{\beta\lambda_k^2}{1+\beta\lambda_k^2}F_k\left(t;u,a\right)\right)\right\}. \tag{18}$$

Substituting the expression $u_k\left(t\right)$ $(k=1,2,...)$ from (15) in (18), we obtain:

$$a(t) = [h(t)]^{-1}\left\{h''(t) \;-f(0,t)+\right.$$

$$+\sum_{k=1}^{\infty}\left(\beta_k^2\left[\left(\int_0^T p_1(t)u_k(t)dt+\varphi_k\right)\cos\beta_k t+\frac{1}{\beta_k}\left(\int_0^T p_2(t)u_k(t)dt+\psi_k\right)\sin\beta_k\;+\right.\right.$$

$$\left.\left.+\;\frac{1}{\beta_k(1+\beta\lambda_k^2)}\int_0^t F_k(\tau\,;u,a)\,\sin\beta_k\left(t-\tau\right)\,d\tau\right]+\frac{\beta\lambda_k^2}{1+\beta\lambda_k^2}F_k(t\,;u,a)\right\}. \tag{19}$$

Thus, the solution of the problem (1)-(3), (7) is reduced to the solution of the system (16),(19) with respect to the unknown functions $u(x,t)$ and $a(t)$.

The following lemma is very important for studying the uniqueness of the solution of problem (1)-(3), (7)

Lemma 2. *If* $\{u(x,t),a(t)\}$ *is any classic solution of problem* (1)-(3), (7)*, the function* $u_k(t)$ $(k=1,2,\ldots)$ *determined by the relation*

$$u_k(t)=2\int_0^1 u(x,t)\cos\lambda_k x\,dx.(k=1,2,....),$$

$[0,T]$ *satisfy on the countable system* (15).

Obviously, if $u_k(t)=2\int_0^1 u(x,t)\cos\lambda_k x dx(k=1,2,...)$ is the solution of the system (15), then the pair $\{u(x,t),a(t)\}$ of functions $u(x,t)=\sum_{k=1}^{\infty}u_k(t)\cos\lambda_k x$ and $a(t)$ is the solution of the system (16), (19) .

The following corollary follows from lemma 2

Corollary 1. *Let the system* (16), (19) *have a unique solution. Then problem* (1)-(3), (7) *can have at most one solution, i.e. if problem* (1)-(3), (7) *has a solution, this solution is unique.*

1. Denote by $B_{2,T}^5$ [14], the totality of all the functions $u(x,t)$ of the form

$$u(x,t)=\sum_{k=1}^{\infty}u_k(t)\cos\lambda_k x\ ,$$

considered in D_T, where each of the functions $u_k(t)$ is continuous on $[0,T]$ and

$$I(u) \equiv \left\{ \sum_{k=1}^{\infty} (\lambda_k^5 \left\| u_k(t) \right\|_{C[0,T]})^2 \right\}^{\frac{1}{2}} < +\infty..$$

We determine the norm in this set as follows:

$$\left\| u(x,t) \right\|_{B_{2,T}^5} = I(u).$$

2. Denote by E_T^5 a space consisting of the topological product

$$B_{2,T}^5 \times C[0,T].$$

The norm of the element $z = \{u, a\}$ is determined by the formula

$$\left\| z \right\|_{E_T^5} = \left\| u(x,t) \right\|_{B_{2,T}^5} + \left\| a(t) \right\|_{C[0,T]}.$$

It is known that $B_{2,T}^5$ and E_T^5 are Banach spaces.

Now in the space E_T^5 we consider the operator

$$\Phi(u,a) = \left\{ \Phi_1(u,a), \Phi_2(u,a) \right\},$$

where

$$\Phi_1(u,a) = \tilde{u}(x,t) = \sum_{k=1}^{\infty} \tilde{u}_k(t) \cos \lambda_k x, \Phi_2(u,a) = \tilde{a}(t),$$

while $\tilde{u}_k(t) (k = 1, 2, ...)$ and $\tilde{a}(t)$ are equal to the right hand sides of (15) and (19) respectively.

It is easy to see that

$$1 + \beta\lambda_k^2 > \beta\lambda_k^2, \quad \frac{1}{1+\beta\lambda_k^2} < \frac{1}{\beta\lambda_k^2},$$

$$\sqrt{\frac{\alpha}{1+\beta}}\,\lambda_k \le \beta_k \le \sqrt{\frac{1+\alpha}{\beta}}\,\lambda_k, \quad \sqrt{\frac{\beta}{1+\alpha}}\,\frac{1}{\lambda_k} \le \frac{1}{\beta_k} \le \sqrt{\frac{1+\beta}{\alpha}}\,\frac{1}{\lambda_k},$$

Taking into account this relation, we find:

$$\left(\sum_{k=1}^{\infty} (\lambda_k^5 \left\| \tilde{u}_k(t) \right\|_{C[0,T]})^2 \right)^{\frac{1}{2}} \le \sqrt{6} \left(\sum_{k=1}^{\infty} (\lambda_k^5 \left| \varphi_k \right|)^2 \right)^{\frac{1}{2}} +$$

$$+\sqrt{6}T \left\| p_1(t) \right\|_{C[0,T]} \left(\sum_{k=1}^{\infty} \left(\lambda_k^5 \left\| u_k(t) \right\|_{C[0,T]} \right)^2 \right)^{\frac{1}{2}} + \sqrt{\frac{6(1+\beta)}{\alpha}} \left(\sum_{k=1}^{\infty} (\lambda_k^4 \left| \psi_k \right|)^2 \right)^{\frac{1}{2}} +$$

$$+\sqrt{\frac{6(1+\beta)}{\alpha}}\,T\left\|p_2(t)\right\|_{C[0,T]}\left(\sum_{k=1}^{\infty}\left(\lambda_k^5\left\|u_k(t)\right\|_{C[0,T]}\right)^2\right)^{\frac{1}{2}}+$$

$$+\frac{1}{\beta}\sqrt{\frac{6(1+\beta)T}{\alpha}}\left(\int_0^T\sum_{k=1}^{\infty}(\lambda_k^2\,|\,f_k(\tau)\,|)^2d\tau\right)^{\frac{1}{2}}+$$

$$\frac{1}{\beta}\sqrt{\frac{6(1+\beta)}{\alpha}}T\left\|a(t)\right\|_{C[0,T]}\left(\sum_{k=1}^{\infty}(\lambda_k^5\left\|u_k(t)\right\|_{C[0,T]})^2\right)^{\frac{1}{2}}, \tag{20}$$

$$\left\|\tilde{a}(t)\right\|_{C[0,T]}\le\left\|[h(t)]^{-1}\right\|_{C[0,T]}\left\{\left\|h''(t)-f(0,t)\right\|_{C[0,T]}+\right.$$

$$+\left(\sum_{k=1}^{\infty}\lambda_k^{-2}\right)^{\frac{1}{2}}\left[\sqrt{\frac{1+\alpha}{\beta}}\left[\left(\sum_{k=1}^{\infty}(\lambda_k^5\,|\varphi_k|)^2\right)^{\frac{1}{2}}+\right.\right.$$

$$+T\left\|p_1(t)\right\|_{C[0,T]}\left(\sum_{k=1}^{\infty}\left(\lambda_k^5\left\|u_k(t)\right\|_{C[0,T]}\right)^2\right)^{\frac{1}{2}}+\sqrt{\frac{1+\beta}{\alpha}}\left(\sum_{k=1}^{\infty}(\lambda_k^4\,|\,\psi_k\,|)^2\right)^{\frac{1}{2}}+$$

$$+\sqrt{\frac{1+\beta}{\alpha}}T\left\|p_2(t)\right\|_{C[0,T]}\left(\sum_{k=1}^{\infty}\left(\lambda_k^5\left\|u_k(t)\right\|_{C[0,T]}\right)^2\right)^{\frac{1}{2}}+$$

$$+\frac{1}{\beta}\sqrt{\frac{T(1+\beta)}{\alpha}}\left(\int_0^T\sum_{k=1}^{\infty}(\lambda_k^2\,|\,f_k(\tau)\,|)^2d\tau\right)^{\frac{1}{2}}+$$

$$\left.+\frac{1}{\beta}\sqrt{\frac{1+\beta}{\alpha}}T\left\|a(t)\right\|_{C[0,T]}\left(\sum_{k=1}^{\infty}(\lambda_k^5\left\|u_k(t)\right\|_{C[0,T]})^2\right)^{\frac{1}{2}}\right]+$$

$$\left.\left.+\left(\sum_{k=1}^{\infty}(\lambda_k^2\,\left\|f_k(t)\right\|_{C[0,T]}|)^2d\tau\right)^{\frac{1}{2}}+\left\|a(t)\right\|_{C[0,T]}\left(\sum_{k=1}^{\infty}(\lambda_k^5\left\|u_k(t)\right\|_{C[0,T]})^2\right)^{\frac{1}{2}}\right]\right\}. \tag{21}$$

Assume that the data of problem (1)-(3), (7) satisfy the following conditions:

1. $\alpha>0,\beta>0,p_1(t)\in C[0,T]\,,p_2(t)\in C[0,T]\,.$

2. $\varphi(x)\in C^4[0,1],\ \varphi^{(5)}(x)\in L_2(0,1),\ \varphi'(0)=\ \varphi(1)=\varphi'''(0)=\varphi''(1)=\varphi^{(4)}(1)=0.$

3. $\psi(x)\in C^3[0,1],\ \psi^{(4)}(x)\in L_2(0,1),\ \psi'(0)=\psi(1)=\psi'''(0)=\psi''(1)=0.$

4. $f(x,t),f_x(x,t)\in C(D_T),\ f_{xx}(x,t)\in L_2(D_T),\ f_x(0,t)=f(1,t)=0\ (0\le t\le T)\,.$

5. $h(t)\in C^2[0,T]\,,\ h(t)\ne 0\ \ (0\le t\le T)$

Then from (20) and (21) we find:

$$\left\{ \sum_{k=1}^{\infty} \left(\lambda_k^5 \parallel \tilde{u}_{2k}(t) \parallel_{C[0\,,T]} \right)^2 \right\}^{\frac{1}{2}} \leq A_1(T) +$$

$$+B_1(T) \left\| a(t) \right\|_{C[0,T]} \left\| u(x,t) \right\|_{B^5_{2,T}} + C_1(T) \left\| u(x,t) \right\|_{B^5_{2,T}} , \tag{22}$$

$$\left\| \tilde{a}(t) \right\|_{C[0,T]} \leq A_2(T) +$$

$$+B_2(T) \left\| a(t) \right\|_{C[0,T]} \left\| u(x,t) \right\|_{B^5_{2,T}} + C_2(T) \left\| u(x,t) \right\|_{B^5_{2,T}} , \tag{23}$$

where

$$A_1(T) = \sqrt{6} \left\| \varphi^{(5)}(x) \right\|_{L_2(0,1)} + \sqrt{\frac{6(1+\beta)}{\alpha}} \left\| \psi^{(4)}(x) \right\|_{L_2(0,1)} +$$

$$+ \frac{1}{\beta} \sqrt{\frac{6T(1+\beta)}{\alpha}} \left\| f_{xx}(x,t) \right\|_{L_2(D_T)} ,$$

$$B_1(T) = \frac{1}{\beta} \sqrt{\frac{6(1+\beta)}{\alpha}} T, C_1(T) = \sqrt{6}\, T \left(\left\| p_1\,(t) \right\|_{C[0,T]} + \sqrt{\frac{1+\beta}{\alpha}} \left\| p_2\,(t) \right\|_{C[0,T]} \right),$$

$$A_2(T) = \left\| [h(t)]^{-1} \right\|_{C[0,T]} \left\{ \left\| h''(t) - f(0,t) \right\|_{C[0,T]} + \right.$$

$$+ \left(\sum_{k=1}^{\infty} \lambda_k^{-2} \right)^{\frac{1}{2}} \left[\sqrt{\frac{1+\alpha}{\beta}} \left[\left\| \varphi^{(5)}(x) \right\|_{L_2(0,1)} + \sqrt{\frac{1+\beta}{\alpha}} \left\| \psi^{(4)}(x) \right\|_{L_2(0,1)} + \right. \right.$$

$$\left. \left. \left. + \frac{1}{\beta} \sqrt{\frac{T(1+\beta)}{\alpha}} \left\| f_{xx}(x,t) \right\|_{L_2(D_T)} \right] + \left\| \left\| f_{xx}(x,t) \right\|_{C[0,T]} \right\|_{L_2(0,1)} \right] \right\},$$

$$B_2(T) = \left\| [h(t)]^{-1} \right\|_{C[0,T]} \left(\sum_{k=1}^{\infty} \lambda_k^{-2} \right)^{\frac{1}{2}} \sqrt{\frac{1+\alpha}{\beta}} \left(\frac{1}{\beta} \sqrt{\frac{1+\beta}{\alpha}} T + 1 \right)$$

$$C_2(T) = \left\| [h(t)]^{-1} \right\|_{C[0,T]} \left(\sum_{k=1}^{\infty} \lambda_k^{-2} \right)^{\frac{1}{2}} \sqrt{\frac{1+\alpha}{\beta}} \left[T \left\| p_1\,(t) \right\|_{C[0,T]} + \right.$$

$$\left. + \sqrt{\frac{1+\beta}{\alpha}} T \left\| p_2\,(t) \right\|_{C[0,T]} \right].$$

Now, from (22)-(23) we obtain:

$$\left\| \tilde{u}(x,t) \right\|_{B^6_{2,T}} + \left\| \tilde{a}(t) \right\|_{C[0,T]} \leq A(T) +$$

$$+B(T) \left\| a(t) \right\|_{C[0,T]} \left\| u(x,t) \right\|_{B^5_{2,T}} + C(T) \left\| u(x,t) \right\|_{B^5_{2,T}} , \tag{24}$$

where

$$A(T) = A_1(T) + A_2(T), B(T) = B_1(T) + B_2(T)\,,$$

$$C(T) = C_1(T) + C_2(T)\,,$$

We now can prove the following theorem

Theorem 2. *If conditions 1- 5 are fulfilled and the inequality*

$$(B(T)(A(T)+2) + C(T))(A(T)+2) < 1\ , \tag{25}$$

is valid, the problem (1)-(3), (7)*) has a unique solution in the sphere* $K = K_R(\|z\|_{E_T^5} \leq R \leq A(T)+2)$ *of the space* E_T^5 .

Proof. In the space E_T^5 we consider the equation

$$z = \Phi z\,, \tag{26}$$

where $z = \{u,\, a\}$, the components $\Phi_i(u,a)$ $(i = 1,2)$ of the operator $\Phi(u,a)$ are determined by the right hand sides of equations (16) and (19).

Let us consider the operator $\Phi(u,a)$ in the sphere $K = K_R$ from E_T^5. Similar to (24) we obtain that for all $z, z_1, z_2 \in K_R$ the following estimations are valid:

$$\|\Phi z\|_{E_T^5} \leq A(T) + B(T)\,\|a(t)\|_{C[0,T]}\,\|u(x,t)\|_{B_{2,T}^5} + C(T)\,\|u(x,t)\|_{B_{2,T}^5} \leq$$

$$\leq A(T) + B(T)(A(T)+2)^2 + C(T)(A(T)+2)\ , \tag{27}$$

$$\|\Phi z_1 - \Phi z_2\|_{E_T^5} \leq B(T)R(\|a_1(t) - a_2(t)\|_{C[0,T]} + \|u_1(x,t) - u_2(x,t)\|_{B_{2,T}^5}) +$$

$$+C(T)\,\|u_1(x,t) - u_2(x,t)\|_{B_{2,T}^5}\ . \tag{28}$$

Now, by virtue of (25), from (27) and (28) it is clear that the operator Φ satisfies the principle of compressed mappings on the set $K = K_R$. Therefore the operator Φ in the sphere $K = K_R$ has a unique fixed point $\{z\} = \{u\,,\,a\}$, that is the solution of equation (26), i.e. it is a unique solution of the system (16), (19) in the sphere $K = K_R$. Then the function $u(x,t)$, as an element of the space $B_{2,T}^5$, is continuous and has continuous derivatives $u_x(x,t)$,$u_{xx}(x,t)$,$u_{xxx}(x,t)$ and $u_{xxxx}(x,t)$ in D_T.

Similar to [13] we can show that $u_t(x,t), u_{tt}(x,t), u_{ttx}(x,t),\ u_{ttxx}(x,t) \in C(D_T)$.

In what follows, we can check that equation (1) and conditions (2), (3), (7) are satisfied in a usual sense. Consequently, $\{u(x,t), a(t)\}$ is the solution of the problem (1)-(3), (7), and by virtue of the Corollary of lemma 2 , it is unique in the sphere $K = K_R$. The theorem is proved.

The validity of the following statement follows directly from theorem 2 and by virtue of theorem 1.

Theorem 3. *Let all the conditions of theorem 2 , and the following agreement conditions be fulfilled*

$$\varphi(0) = h(0) \ - \int_0^T p_1(t)h(t)dt, \quad \psi(0) = h''(0) - \int_0^T p_2(t)h(t)dt$$

and

$$\left(T \, \|p_2(t)\|_{\ C[0,T]} + \|p_1(t)\|_{\ C[0,T]} + \frac{T}{2}(A(T)+2) \right) T < 1$$

Then problem (1)-(4) *has a classic unique solution in the sphere* $K = K_R(\|z\|_{E_T^5} \leq R = A(T) + 2)$ *of the space* E_T^5.

References

[1] S.D. Algazin, I.A. Kiyko, *Flutter of plates and shells*, Moscow, Nauka, 2006, 248 p. (in Russian).

[2] S.A. Shabrov, *Estimates of the impact function of a mathematical model of the fourth-order*, Vestnik Voronezhskogo Gosudarstvennogo Universiteta: Seriya. Fizika-Matematika, **2**, 2015, 168-179 (in Russian).

[3] D.J. Benney, J.C. Luke, *On the interactions of permanent waves of nite amplitude*, Journal of Mathematical Physics, **43**, 1964, 309–319. doi:10.1002/sapm1964431309.

[4] A.N. Tikhonov, *On the stability of inverse problems*, Doklady of the USSR Academy of Science, **39**, 1943, 195-198.

[5] M.M. Lavrentiev, On one inverse problem for wave equation, Dokl. AN SSSR, **157(3)**, 1964, 520-521 (in Russian).

[6] M.M. Lavrentyev, V.G. Romanov, S.T. Shishatskii, *Ill-posed Problems of Mathematical Physics and Analysis*, Moscow, Nauka, 1980 288 p. (in Russian).

[7] V.K. Ivanov, V.V. Vasin, V.P. Tanina, *Theory of linear ill-posed problems and its applications*, Moscow, Nauka, 1978, 206 p.

[8] A.M. Denisov, *Introduction to the theory of inverse problems*, Moscow, Moscow State University, 1994, 206 p.

[9] T.K. Yuldashev, *On a nonlocal inverse problem for a Benney-Luke type integro-differential equation with degenerate kernel*, Vestnik TVGU Seriya Prikladnaya Matematika, **3**, 19–41, 2018 (in Russian).

[10] Ya.T. Mehraliev, B.K. Valiyeva, *Inverse boundary value problem for the linearized Benny-Luke equation with nonlocal conditions*, Vestn. Udmurtsk un-ta. Math. Fur. Computer. Sciences, **29(2)**, 2019, 166–182.

[11] Ya.T. Mehraliyev, B.K. Valiyeva, *On one nonlocal inverse boundary problem for the Benney –Luke equation with integral conditions*, IOP Conf. Series: Journal of Physics: Conf. Series 1203 (2019) 012100 IOP Publishing. doi:10.1088/1742-6596/1203/1/012100

[12] Ya.T. Mehraliyev, B.K. Valiyeva, A.T. Ramazanova, *An inverse boundary value problem for a linearized Benny–Luke equation with nonlocal boundary conditions*, Cogent Mathematics & Statistics, **6(1)**, 2019, 1634316.

[13] Ya.T. Mehraliyev, F.Kh. Alizade, *An inverse boundary value problem for a fourth-order Boussinesq equation with nonlocal time-integral conditions of the second kind*, Vestn. Udmurtsk un-ta. Math. Fur. Computer. Sciences, **26(4)**, 2016, 503–514.

[14] K.I. Khudaverdiyev, A.A. Veliyev, *Investigation of a one-dimensional mixed problem for a class of pseudohyperbolic equations of third order with non-linear operator right hand side*, Baku, Chashyoghly, 2010, 168 p. (in Russian).

Veliyeva Bahar Kamal kizi
Ganja State University
E-mail: bahar.veliyeva.91@inbox.ru

Received 24 September 2023
Accepted 13 November 2023

Caspian Journal of Applied Mathematics, Ecology and Economics
V. 11, No 2, 2023, December
ISSN 1560-4055

Global Bifurcation from Infinity of Nondifferentiable Perturbations of Half-linear Eigenvalue Problems for Ordinary Differential Equations of Fourth Order

M.M. Mammadova

Abstract. In this paper, we consider the nondifferentiable perturbations of a half-linear eigenvalue problem for ordinary differential equations of fourth order. We show the existence of global components of a set of nontrivial solutions bifurcating from intervals containing half-eigenvalues of a half-linear eigenvalue problem and contained in classes of functions that have oscillatory properties of half-eigenfunctions of a half-linear eigenvalue problem and their derivatives in some neighborhoods of these intervals.

Key Words and Phrases: nondifferentiable perturbations, half-linear eigenvalue problem, half-eigenfunction, bifurcation interval, global component

2010 Mathematics Subject Classifications: 34B05, 34B24, 34C23, 34L20, 34l30, 34K18, 47J10, 47J15

1. Introduction

We consider the following nonlinear eigenvalue problem

$$\begin{aligned} \ell y \equiv (p\,(x)\,y'')'' - (q(x)y')' + r(x)y = \lambda\tau(x)\,y + \alpha(x)y^{+}(x) + \beta(x)y^{-}(x) + \\ f(x,y,y',y'',y''',\lambda) + g(x,y,y',y'',y''',\lambda),\ x\in(0,l), \end{aligned} \tag{1.1}$$

$$y(0)=y'(0)=y(l)=y'(l)=0,, \tag{1.2}$$

where $\lambda\in\mathbb{R}$ is a parameter, $p\in C^2\,([0,l];(0,+\infty))$, $q\in C^1\,([0,l];[0,+\infty))$, $\tau\in C\,([0,l];(0,+\infty))$, r, α, $\beta\in C\,([0,l];\mathbb{R})$, and $y^+=\max\,\{y,0\}$, $y^-=(-y)^+$. The nonlinear terms f and g satisfy the following conditions: f, $g\in C\left([0,l]\times\mathbb{R}^5;\mathbb{R}\right)$; there exists constant $M>0$ such that

$$\left|\frac{f(x,y,s,v,w,\lambda)}{y}\right| \le M,\ x\in[0,l],\ (y,s,v,w)\in\mathbb{R}^4,\ y\neq0,\ \lambda\in\mathbb{R}; \tag{1}$$

$$g(x,y,s,v,w,\lambda)=o(|y|+|s|+|v|+|w|)\ \text{as}\ |y|+|s|+|v|+|w|\to+\infty, \tag{2}$$

uniformly in $x \in [0, l]$ and in $\lambda \in \Lambda$, for any bounded interval $\Lambda \subset \mathbb{R}$.

The global bifurcation of nontrivial solutions from zero and infinity to nonlinear Sturm-Liouville problems of second and fourth orders was studied in detail in [1, 3, 5, 10, 11, 13, 14] (see also references therein). These papers prove the existence of global continua of nontrivial solutions in $\mathbb{R} \times C^1$ and $\mathbb{R} \times C^3$, respectively, emanating from bifurcation points and bifurcation intervals (in $\mathbb{R} \times \{0\}$ and $\mathbb{R} \times \{\infty\}$, respectively, which we identify with $\mathbb{R}$), containing the eigenvalues of linear problems obtained from nonlinear problems by setting the nonlinear terms to zero. Moreover, it has been established that these global continua in the neighborhoods of bifurcation points and intervals are contained in classes of functions with fixed oscillation count.

Global bifurcation from zero and infinity in half-linearizable Sturm-Liouville problems of second order was studied in [5, 12, 15] (it should be noted that the nonlinear problems considered in [15] are nonlinear perturbations of linear eigenvalue problems for self-adjoint operators of $2m$th-order). These papers show the existence of global continua of nontrivial solutions in $\mathbb{R} \times C^1$, bifurcating from half-eigenvalues of half-linear eigenvalue problems and contained in classes of functions having the usual nodal properties. Similar results for half-linearizable Sturm-Liouville problems of fourth order were obtained in [2, 7, 8, 15].

In [9] we consider the nondifferentiable perturbations of a half-linear Sturm-Liouville problem of fourth order. We prove the existence of two families of global components of the set of nontrivial solutions in $\mathbb{R} \times C^3$, emanating from bifurcation intervals containing the half-eigenvalues of the corresponding half-linear eigenvalue problem and contained in classes of functions having oscillation properties of half-eigenfunctions (and their derivatives) of this half-linear eigenvalue problem.

In this paper, we consider global bifurcation from infinity of nonlinear eigenvalue problem (1.1), (1.2) under conditions (1.3) and (1.4). We prove the existence of two families unilateral global components of a set of nontrivial solutions bifurcating from intervals containing half-eigenvalues of a half-linear eigenvalue problem (1.1), (1.2) with $f, g \equiv 0$ and contained in classes of functions that have oscillatory properties of half-eigenfunctions of this half-linear eigenvalue problem and their derivatives in some neighborhoods of these intervals.

2. Preliminary

Let E be the Banach space

$$C^3[0, l] \cap \{y : y(0) = y'(0) = y(1) = y'(1) = 0\}$$

with the usual norm

$$||y||_3 = ||y||_\infty + ||y'||_\infty + ||y''||_\infty + ||y'''||_\infty,$$

where $||y||_\infty = \max\limits_{x \in [0, l]} |y(x)|$.

Using an extension of the Prüfer type transformation (see [4]), in paper [1] were constructed the sets S_k^ν, $k \in \mathbb{N}$, $\nu \in \{+, -\}$, of functions of E that have the oscillation properties of eigenfunctions of the linear eigenvalue problem

$$\begin{cases} (\ell y)(x) = \lambda \tau(x)\, y(x), \ x \in (0,l), \\ y(0) = y'(0) = y(l) = y'(l) = 0, \end{cases}$$

and their derivatives [1,§ 3.1].

It follows from [15, Theorem 3.3] that half-linear eigenvalue problem

$$\begin{cases} (\ell y)(x) = \lambda \tau(x)\, y(x) \ + \alpha(x) y^+(x) \ + \beta(x) y^-(x), \ x \in (0,l), \\ y(0) = y'(0) = y(l) = y'(l) = 0, \end{cases} \tag{2.1}$$

has two infinitely increasing sequences $\{\lambda_k^+\}_{k=1}^\infty$ and $\{\lambda_k^-\}_{k=1}^\infty$ of simple half-eigenvalues such that

$$\text{if } k' > k \geq 1, \text{ then } \lambda_{k'}^{\nu'} > \lambda_k^\nu \text{ for each } \nu,\, \nu' \in \{+, -\}.$$

Moreover, for each $k \in \mathbb{N}$ and each $\nu \in \{+, -\}$ the half-eigenfunction $y_{k,\nu}$ corresponding to the half eigenvalue λ_k^ν of problem (2.1) lies in S_k^ν.

We introduce the following notations (see [9, p. 30]):

$$N_\alpha = \max_{x \in [0,\, l]} |\alpha(x)|, \ N_\beta = \max_{x \in [0,\, l]} |\beta(x)|, \ \tau_0 = \min_{x \in [0,\, l]} \tau(x),$$

$$I_k^+ = \left[\lambda_k^+ - \frac{N_\alpha + N_\beta + M}{\tau_0}, \lambda_k^+ + \frac{N_\alpha + N_\beta + M}{\tau_0}\right],$$

$$I_k^- = \left[\lambda_k^- - \frac{N_\alpha + N_\beta + M}{\tau_0}, \lambda_k^- + \frac{N_\alpha + N_\beta + M}{\tau_0}\right].$$

Remark 2.1. If the function $g \in C\left([0,l] \times \mathbb{R}^5; \mathbb{R}\right)$ satisfies the condition

$$g(x,y,s,v,w,\lambda) = o(|y| + |s| + |v| + |w|) \text{ as } |y| + |s| + |v| + |w| \to 0, \tag{2.2}$$

uniformly in $x \in [0,l]$ and in $\lambda \in \Lambda$, for any bounded interval $\Lambda \subset \mathbb{R}$, then we consider the bifurcation from zero, i.e. the existence of solutions to problem (1.1), (1.2) having an arbitrarily small norm. Note that the bifurcation from zero of nontrivial solutions to problem (1.1), (1.2) was studied in [2] in the case of $f \equiv 0$, and in [9] in the case of $f \not\equiv 0$. These papers studied not only the existence of nontrivial solutions with sufficiently small norms, but also the existence of nontrivial solutions with sufficiently large norms. In the case of $f \equiv 0$ it was shown that for each $k \in \mathbb{N}$ and each $\nu \in \{+, -\}$ there exists a continuum C_k^ν of solutions of problem (1.1), (1.2) that contains $(\lambda_k^\nu, 0)$, contained in $(\mathbb{R} \times S_k^\nu) \cup \{(\lambda_k^\nu, 0)\}$ and unbounded in $\mathbb{R} \times E$. In the case of $f \not\equiv 0$ it was proven that for each $k \in \mathbb{N}$ and each $\nu \in \{+, -\}$ there exists a component D_k^ν of solutions of problem (1.1), (1.2) that contains $I_k^\nu \times \{0\}$, contained in $(\mathbb{R} \times S_k^\nu) \cup \{(I_k^\nu \times \{0\})\}$ and unbounded in $\mathbb{R} \times E$.

Remark 2.2. Since condition (1.4) is satisfied, we consider the bifurcation from infinity, i.e., the existence of solutions to problem (1.1), (1.2) having an arbitrarily large norm. If, in addition (1.4), $f \equiv 0$, then the problem is called asymptotically linear and the existence of solutions with sufficiently large norms bifurcating from infinity is discussed in [8]. We recall that in [8] for each $k \in \mathbb{N}$ and each $\nu \in \{+, -\}$ it is established the existence of a continuum C_k^ν of nontrivial solutions to problem (1.1), (1.2) with $f \equiv 0$ containing (λ_k^ν, ∞) and neighborhood Q_k^ν of the point (λ_k^ν, ∞) such that $(\mathcal{C}_k^\nu \cap \mathcal{Q}_k^\nu) \subset \mathbb{R} \times S_k^\nu$ and either $\mathcal{C}_k^\nu$ meets $(\lambda_{k'}^{\nu'}, \infty)$ with respect to the set $\mathbb{R} \times S_{k'}^{\nu'}$ for some $(k', \nu') \neq (k, \nu)$, or $\mathcal{C}_k^\nu \backslash Q_k^\nu$ meets $\{(\lambda, 0) : \lambda \in \mathbb{R}\}$ for some $\lambda \in \mathbb{R}$, or the natural projection $P_\mathbb{R}(\mathcal{C}_k^\nu)$ of the set $\mathcal{C}_k^\nu$ onto $\mathbb{R} \times \{0\}$ is unbounded. If $f \not\equiv 0$, then in next section we show that the bifurcation occurs from some intervals of the line $\mathbb{R} \times \{\infty\}$. Moreover, we will study global character of the union of all components of the set of nontrivial solutions bifurcating from these intervals.

3. Global bifurcation from infinity of nontrivial solutions of problem (1.1), (1.2)

To study the global bifurcation of solutions to problem (1.1), (1.2) with condition (1.4) we will use the approach used in [3, 11, 13], which consists in transforming the problem of bifurcation from infinity to a problem involving bifurcation from zero.

Let $\mathcal{C}$ be the set of nontrivial solutions of problem (1.1), (1.2). If $(\lambda, y) \in \mathcal{C}$, then dividing (1.1), (1.2) by $||y||_3^2$ and setting $v = \frac{y}{||y||_3^2}$ we get

$$\begin{cases} \ell(v) = \lambda\tau(x)v + \alpha v^+ + \beta v^- + \frac{f(x, y, y', y'', y''', \lambda)}{||y||_3^2} + \frac{g(x, y, y', y'', y''', \lambda)}{||y||_3^2}, \ 0 < x < l, \\ v(0) = v'(0) = v(l) = v'(l) = 0. \end{cases} \tag{3.1}$$

Note that $||v||_3 = \frac{1}{||y||_3}$, and consequently, $||y||_3 = \frac{1}{||v||_3}$ and $y = \frac{v}{||v||_3^2}$. We define the functions $\hat{f}(\lambda, v)$, $\hat{g}(\lambda, v) \in C[0, l]$ as follows:

$$\hat{f}(\lambda, v)(x) = \begin{cases} ||v||_3^2 f\left(x, \frac{v(x)}{||v||_3^2}, \frac{v'(x)}{||v||_3^2}, \frac{v''(x)}{||v||_3^2}, \frac{v'''(x)}{||v||_3^2}, \lambda\right) & \text{if } v(x) \neq 0, \\ 0 & \text{if } v(x) = 0, \end{cases} \tag{3.2}$$

$$\hat{g}(\lambda, v)(x) = \begin{cases} ||v||_3^2 g\left(x, \frac{v(x)}{||v||_3^2}, \frac{v'(x)}{||v||_3^2}, \frac{v''(x)}{||v||_3^2}, \frac{v'''(x)}{||v||_3^2}, \lambda\right) & \text{if } v(x) \neq 0, \\ 0 & \text{if } v(x) = 0. \end{cases} \tag{3.3}$$

Then problem (3.1) reduces to the following equivalent form

$$\begin{cases} \ell(v) = \lambda\tau(x)v + \alpha(x)v^+ + \beta(x)v^- + \hat{f}(\lambda, v) + \hat{g}(\lambda, v), \ 0 < x < l, \\ v(0) = v'(0) = v(l) = v'(l) = 0. \end{cases} \tag{3.4}$$

Obviously, for each $\lambda \in \mathbb{R}$ the pair $(\lambda, 0)$ is a trivial solution to problem (3.4). By (3.2) it follows from conditions (1.3) that

$$||\hat{f}(\lambda, v)||_\infty = ||v||_3^2 \left\| f\left(x, \frac{v}{||v||_3^2}, \frac{v'}{||v||_3^2}, \frac{v''}{||v||_3^2}, \frac{v'''}{||v||_3^2}, \lambda\right) \right\|_\infty \leq$$
$$M||v||_3^2 \left\| \frac{v}{||v||_3^2} \right\|_\infty = M \, \|v\|_\infty \,, \tag{3.5}$$

In view of condition (1.4), by [1, Lemma 5.5] for any sufficiently small $\varepsilon > 0$ there exists sufficiently large $\Delta_\varepsilon > 0$ such that

$$|g(x, y, y', y'', y''', \lambda)| < \varepsilon ||y||_3 \text{ for } \lambda \in \Lambda \text{ and } ||y||_3 > \Delta_\varepsilon. \tag{3.6}$$

Let $\delta_\varepsilon = \frac{1}{\Delta_\varepsilon}$ and $\lambda \in \Lambda$, $||v||_3 < \delta_\varepsilon$. Then we have $||y||_3 = \frac{1}{||v||_3} > \frac{1}{\delta_\varepsilon} = \Delta_\varepsilon$, and consequently, by (3.6) we get

$$\frac{||\hat{g}(\lambda, v)||_\infty}{||v||_3} = \frac{||v||_3^2 \left\| g\left(x, \frac{v}{||v||_3^2}, \frac{v'}{||v||_3^2}, \frac{v''}{||v||_3^2}, \frac{v'''}{||v||_3^2}, \lambda\right) \right\|_\infty}{||v||_3} =$$

$$\frac{\left\| g\left(x, \frac{v}{||v||_3^2}, \frac{v'}{||v||_3^2}, \frac{v''}{||v||_3^2}, \frac{v'''}{||v||_3^2}, \lambda\right) \right\|_\infty}{\frac{1}{||v||_3}} = \frac{\|g(x, y, y', y'', y''', \lambda)\|_\infty}{||y||_3} < \varepsilon,$$

which implies that

$$||\hat{g}(\lambda, \, v)||_\infty = o(||v||_3), \quad \text{as } ||v||_3 \to 0, \tag{3.7}$$

uniformly for $\lambda \in \Lambda$.

Thus by (3.5) and (3.7) the transformation

$$H : (\lambda, y) \to (\lambda, v)$$

turns a "bifurcation at infinity" problem (1.1), (1.2) into a "bifurcation from zero" problem (3.4).

Let $\hat{\mathcal{C}}$ we denote the set of nontrivial solutions to problem (3.4). By construction, the transformation H maps $\mathcal{C}$ into $\hat{\mathcal{C}}$ and, heuristically, interchanges points at $y = 0$ (respectively, $y = \infty$) with points at $y = \infty$ (respectively, $y = 0$).

Remark 3.1. It follows from [9, Lemma 2.2] that for each $k \in \mathbb{N}$ and each $\nu \in \{+, -\}$ the set of bifurcation points to problem (3.4) with respect to the set $\mathbb{R} \times S_k^\nu$ is nonempty. Moreover, if $(\lambda, 0)$ is a bifurcation point of this problem with respect to the set $\mathbb{R} \times S_k^\nu$, then $\lambda \in I_k^\nu$.

Remark 3.2. By the above arguments it follows from Remark 3.1 that for each $k \in \mathbb{N}$ and each $\nu \in \{+, -\}$ the set of bifurcation points to problem (1.1), (1.2) with respect to the set $\mathbb{R} \times S_k^\nu$ is nonempty. Moreover, if (λ, ∞) is a bifurcation point of this problem with respect to the set $\mathbb{R} \times S_k^\nu$, then $\lambda \in I_k^\nu$.

For each $k \in \mathbb{N}$ and each $\nu \in \{+, -\}$ by $\hat{\mathcal{D}}_k^\nu$ we define the union of all the components of $\hat{\mathcal{C}}$ which meet $I_k^\nu \times \{0\}$ with respect to $\mathbb{R} \times S_k^\nu$. Using Remark 3.1 and by following the arguments of Theorem 3.1 on p. 151 of [13] we can show that $\hat{\mathcal{D}}_k^\nu$ is not empty.

Remark 3.3. Since condition (1.4) is satisfied, statement [1, Lemma 1.1] does not hold. For this reason, for problem (3.4), statement [9, Theorem 3.1] does not hold.

However, by extending the approximation technique from [5] and combining it with the global bifurcation results in [6], [9], [10] and [14] we can show that for each $k \in \mathbb{N}$ and each $\nu \in \{+, -\}$ for $\hat{\mathcal{D}}_k^\nu$ the following statements hold:

(a) there exists a neighborhood $\hat{\mathcal{Q}}_k^\nu$ of $I_k^\nu \times \{0\}$ in $\mathbb{R} \times E$ such that
$\hat{\mathcal{D}}_k^\nu \bigcap \hat{\mathcal{Q}}_k^\nu \subset \mathbb{R} \times S_k^\nu$;

(b) either $\hat{\mathcal{D}}_k^\nu \bigcap \hat{\mathcal{D}}_{k'}^{\nu'} \neq \emptyset$ for some $(k', \nu') \neq (k, \nu)$, or $\hat{\mathcal{D}}_k^\nu$ is unbounded in $\mathbb{R} \times E$; if in this case $\hat{\mathcal{D}}_k^\nu$ is unbounded in $\mathbb{R} \times E$, then either $\hat{\mathcal{D}}_k^\nu$ meets $\mathbb{R} \times \{\infty\}$ for some $\lambda \in \mathbb{R}$, or natural projection $P_{\mathbb{R}}(\hat{\mathcal{D}}_k^\nu)$ of $\hat{\mathcal{D}}_k^\nu$ onto $\mathbb{R} \times \{0\}$ is unbounded.

Now for each $k \in \mathbb{N}$ and each $\nu \in \{+, -\}$ by $\mathcal{D}_k^\nu$ we define the union of all the components of $\mathcal{C}$ which meet $I_k^\nu \times \{\infty\}$ with respect to $\mathbb{R} \times S_k^\nu$. It is obvious that $\mathcal{D}_k^\nu$ is not empty and $\mathcal{D}_k^\nu$ is the inverse image $H^{-1}(\hat{\mathcal{D}}_k^\nu)$ of $\hat{\mathcal{D}}_k^\nu$ under the transformation H. Then, using statements (a) and (b) for the set $\hat{\mathcal{D}}_k^\nu$ and Remark 3.2, we obtain the following result on the global bifurcation from infinity for problem (1.1), (1.2).

Theorem 3.1. *For each $k \in \mathbb{N}$ and each $\nu \in \{+, -\}$ the set $\mathcal{D}_k^\nu$ has the following properties:*

(i) there exists a neighborhood $\mathcal{Q}_k^\nu$ of $I_k^\nu \times \{\infty\}$ in $\mathbb{R} \times E$ such that
$\mathcal{D}_k^\nu \cap \mathcal{Q}_k^\nu \subset \mathbb{R} \times S_k^\nu$;

(ii) either $\mathcal{D}_k^\nu \cap \mathcal{D}_{k'}^{\nu'} \neq \emptyset$ for some $(k', \nu') \neq (k, \nu)$, or $\mathcal{D}_k^\nu$ meets $\mathbb{R} \times \{0\}$ for some $\lambda \in \mathbb{R}$, or natural projection $P_{\mathbb{R}}(\mathcal{D}_k^\nu)$ of $\mathcal{D}_k^\nu$ onto $\mathbb{R} \times \{0\}$ is unbounded.

Next, if both conditions (1.4) and (2.3) are satisfied, then we can improve Remark 2.1 and Theorem 3.1 as follows.

Theorem 3.2. *If conditions (1.4) and (2.3) are satisfied, then for each $k \in \mathbb{N}$ and each $\nu \in \{+, -\}$ we have*

$$\mathcal{D}_k^\nu \subset \mathbb{R} \times S_k^\nu \text{ and } \mathcal{D}_k^\nu \cap \mathcal{D}_{k'}^{\nu'} = \emptyset \text{ for any } (k', \nu') \neq (k, \nu).$$

Moreover, if $\mathcal{D}_k^\nu$ meets $\mathbb{R} \times \{0\}$ for some $\lambda \in \mathbb{R}$, then $\lambda \in I_k^\nu$. In a similar way, if $\hat{\mathcal{D}}_k^\nu$ meets $\mathbb{R} \times \{\infty\}$ for some $\lambda \in \mathbb{R}$, then $\lambda \in I_k^\nu$.

The proof of this theorem is similar to that of [13, Theorem 3.3]

References

[1] Z.S. Aliev, On the global bifurcation of solutions of some nonlinear eigenvalue problems for ordinary differential equations of fourth order, *Sb. Math.*, **207**(12) (2016), 1625-1649.

[2] Z.S. Aliev, M.M. Mammadova, On global bifurcation of one half-linearizable eigenvalue problem, *News BSU. Ser. Phys.-Math. Sci.*, (1) (2021), 29-37.

[3] Z.S. Aliyev, N.A. Mustafayeva, Bifurcation of solutions from infinity for certain nonlinear eigenvalue problems of fourth-order ordinary differential equations, *Electron. J. Differ. Equ.*, **2018**(98) (2018), 1-19.

[4] D.O. Banks, G.J. Kurowski, Prüfer transformation for the equation of a vibrating beam subject to axial forces, *J. Differential Equations*, **24**(1) (1977), 57-74.

[5] H. Berestycki, On some nonlinear Sturm-Liouville problems, *J. Differential Equations*, **26**(3) (1977), 375-390.

[6] E.N. Dancer, On the structure of solutions of non-linear eigenvalue problems, *Indiana Univ. Math. J.*, **23**(11) (1974), 1069-1076.

[7] A.C. Lazer, P.J. McKenna, Global bifurcation and a theorem of Tarantello, *J. Math. Anal. Appl.*, **181**(3) (1994), 648-655.

[8] M.M. Mammadova, On some asymptotically half-linear eigenvalue problem for ordinary differential equations of fourth order, *Proc. IMM*, **48**(1) (2022), 113-122.

[9] M.M. Mammadova, Global bifurcation from zero in nondifferentiable perturbations of half-linear fourth-order eigenvalue problems, *Proc. IMM*, **49**(1) (2023), 28-37.

[10] P.H. Rabinowitz, Some global results for nonlinear eigenvalue problems, *J. Funct. Anal.*, **7**(3) (1971), 487-513.

[11] P.H. Rabinowitz, On bifurcation from infinity, *J. Differential Equations*, **14**(3) (1973), 462-475.

[12] B.P. Rynne, Bifurcation from infinity in nonlinear Sturm-Liouville problems with different linearizations at $u = \pm\infty$, *Appl. Anal.*, **67**(3-4) (1997), 233-244.

[13] B.P. Rynne, Bifurcation from zero or infinity in Sturm-Liouville problems which are not linearizable, *J. Math. Anal. Appl.*, **228**(1) (1998), 141-156.

[14] B.P. Rynne, Infinitely many solutions of superlinear fourth order boundary value problems, *Topol. Methods Nonlinear Anal.*, **19**(2) (2002), 303-312.

[15] B.P. Rynne, Half-eigenvalues of self-adjoint, $2m$th-order differential operators and semilinear problems with jumping nonlinearities, *Diff. Int. Equat.*, **14**(9) (2001), 1129-1152.

Masuma M. Mammadova
Baku State University, Baku, Azerbaijan
E-mail: memmedova.mesume@inbox.ru

Received 16 September 2023
Accepted 17 November 2023

Caspian Journal of Applied Mathematics, Ecology and Economics
V. 11, No 2, 2023, December
ISSN 1560-4055

Global Bifurcation from Infinity in Nonlinear Dirac Problems with Eigenvalue Parameter in the Boundary Conditions

Nigar S. Aliyeva

Abstract. In this paper we consider global bifurcation from infinity for nonlinear Dirac system with a spectral parameter in boundary conditions. We show the existence of two families of global continua of the set of solutions to this problem, emanating from the points of the line $\mathbb{R} \times \{\infty\}$, corresponding to the eigenvalues of the linear problem, which is obtained from the nonlinear problem by setting the nonlinear term to zero and contained in classes of vector-functions with a fixed oscillation count in the neighborhood of these points.

Key Words and Phrases: nonlinear Dirac problem, eigenvalue, eigenvector-function, bifurcation from infinity, asymptotic bifurcation point, oscillation count

2010 Mathematics Subject Classifications: 34A30, 34B05, 34B15, 34C10, 34C23, 34K29, 47J10, 47J15

1. Introduction

We consider the following nonlinear Dirac equation

$$(\ell w)(x) \equiv Bw'(x) - P(x)w(x) = \lambda w(x) + g(x, w(x), \lambda),\ 0 < x < \pi, \tag{1.1}$$

subject to the boundary conditions

$$\begin{aligned} U_1(\lambda, w) := (\lambda \cos\alpha + a_0, \lambda \sin\alpha + b_0)w(0) = \\ (\lambda \cos\alpha + a_0)\, v(0) + (\lambda \sin\alpha + b_0)\, u\,(0) = 0, \end{aligned} \tag{1.2}$$

$$\begin{aligned} U_2(\lambda, w) := (\lambda \cos\beta + a_1, \lambda \sin\beta + b_1)w(\pi) = \\ (\lambda \cos\beta + a_\pi)\, v(0) + (\lambda \sin\beta + b_0)\, u\,(\pi) = 0, \end{aligned} \tag{1.3}$$

where

$$B = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},\ P(x) = \begin{pmatrix} p\,(x) & 0 \\ 0 & r(x) \end{pmatrix},\ w(x) = \begin{pmatrix} u(x) \\ v(x) \end{pmatrix},$$

$\lambda \in \mathbb{C}$ is an eigenvalue parameter, $p(x)$, $r(x) \in C([0,\pi];\mathbb{R})$, $[0,\ \pi]$, α, β, a_0, b_0, a_1 and b_1 are real constants such that $0 \leq \alpha,\ \beta < \pi$ and

$$\sigma_0 = a_0 \sin\alpha - b_0 \cos\alpha < 0,\ \sigma_1 = a_1 \sin\beta - b_1 \cos\beta > 0. \tag{1.4}$$

Here the function $g = \begin{pmatrix} g_1 \\ g_2 \end{pmatrix} \in C\left([0,\pi] \times \mathbb{R}^2 \times \mathbb{R}; \mathbb{R}^2\right)$ satisfies the following condition:

$$g(x, w, \lambda) = o(|w|) \text{ as } |w| \to \infty, \tag{1.5}$$

uniformly in $(x, \lambda) \in [0,\pi] \times \Lambda$, for any bounded interval $\Lambda \subset \mathbb{R}$ (we denote by $|\cdot|$ the Euclidean norm in $\mathbb{R}^2$).

In relativistic quantum mechanics and particle physics, the Dirac equation is a relativistic wave equation for describing spin-1/2 particles (i.e., fermions) [13, 19]. This equation is consistent with both the principles of quantum mechanics and the special theory of relativity, and was the first theory to fully explain the theory of relativity in the context of quantum mechanics.

The nonlinear Dirac equation since the first nonlinear generalization of the linear Dirac equation by the Russian physicist D.D. Ivanenko [11] naturally emerged as a practical model in many physical systems. Note that the nonlinear Dirac equations model the state of relativistic electrons (or other 1/2 particles) in external fields and during self-interaction. For example, equation (1.1) in the case of $p(x) = V(x) + m$, $r(x) = V(x) - m$ and $g(x, w, \lambda) = \tilde{g}(x, |w|)w$, where $V(x)$ is the potential function, m is the particle mass and $\tilde{g}$ is a nonlinear self-coupling, describes the dynamics of self-acting massive Dirac fermions with spin 1/2 inside external non-stationary electromagnetic potentials [19].

The global bifurcation from infinity in nonlinear Sturm-Liouville problems, fourth-order nonlinear Sturmian systems and nonlinear Dirac systems in cases where the spectral parameter is not contained in the boundary conditions and is contained in the boundary condition has been well studied in [3-5, 9, 14-18, 20]. In these papers, the authors show the existence of global connected components of the set of solutions to the nonlinear problems under consideration, which branch from the bifurcation points and intervals of the line $\mathbb{R} \times \{\infty\}$. Moreover, these components are contained in classes of functions that have oscillating properties of eigenvector-functions of linear problems, obtained from nonlinear problems by equating nonlinear terms to zero, in the neighborhoods of bifurcation points and intervals.

Note that global bifurcation from zero of nontrivial solutions to the nonlinear Dirac problem (1.1)-(1.3) was studied in [2].

In this paper, we consider the structure and behavior of global continua bifurcating from infinity to the nonlinear Dirac problem (1.1)-(1.3).

The rest of the article is organized as follows. Section 2 first provides information about the oscillatory properties of the eigenvector functions of the linear problem (1.1)-(1.3) for $g \equiv 0$ in terms of the Prüfer angular function (see [8]). Then we present the construction of classes of vector-functions in $C([0,\pi];\mathbb{R}^2)$ and $C([0,\pi];\mathbb{R}^2) \oplus \mathbb{R}^2$, that have these oscillatory properties of these eigenvector-functions. Moreover, the problem is reduced to equivalent operator equations in the space $C([0,\pi];\mathbb{R}^2\oplus\mathbb{R}^2)$. Section 3 shows the existence of global continua of nontrivial solutions to problem (1.1)-(1.3), emanating from asymptotic bifurcation points and contained in the above-mentioned classes of vector-functions in the neighborhoods of these points.

2. Preliminary

Let the continuous functions $\gamma(\lambda)$ and $\delta(\lambda)$ on $\mathbb{R}$ be defined by

$$\cot\gamma(\lambda) = -\frac{\lambda\cos\alpha + a_0}{\lambda\sin\alpha + b_0},\ \gamma\left(-\frac{b_0}{\sin\alpha}\right) = 0,\ \text{for } \alpha\neq 0; \tag{2.1}$$

$$\cot\delta(\lambda) = -\frac{\lambda\cos\beta + a_1}{\lambda\sin\beta + b_1},\ \delta\left(-\frac{b_0}{\sin\beta}\right) = 0,\ \text{for } \beta\neq 0. \tag{2.2}$$

In view of (1.4) the function $\gamma(\lambda)$ strictly increases and the function $\delta(\lambda)$ strictly decreases in the domains of their definition. Moreover, it follows from (2.1) and (2.2) that

$$\gamma(\lambda)\in(-\alpha,\pi-\alpha),\ \lim_{\lambda\to-\infty}\gamma(\lambda) = -\alpha \ \text{ and } \ \lim_{\lambda\to+\infty}\gamma(\lambda) = \pi-\alpha, \tag{2.3}$$

$$\delta(\lambda)\in(-\beta,\pi-\beta),\ \lim_{\lambda\to-\infty}\delta(\lambda) = \pi-\beta \ \text{ and } \ \lim_{\lambda\to+\infty}\delta(\lambda) = -\beta. \tag{2.4}$$

Let $E = C\left([0,\pi];\,\mathbb{R}^2\right)$ be the Banach space with the usual norm

$$||w|| = \max_{x\in[0,\,\pi]}|u(x)| + \max_{x\in[0,\,\pi]}|v(x)|.$$

Let S be the subset of E defined as follows:

$$S = \{w\in E:\ |u(x)+|v(x)|\ > 0,\ x\in[0,\pi]\}\,.$$

For each fixed $(\lambda, w) = \left(\lambda, \begin{pmatrix} u \\ v \end{pmatrix}\right)\in\mathbb{R}\times S$ we define a continuous function $\theta(\lambda,w,x)$ on $[0,\pi]$ by

$$\cot\theta(\lambda,w,x) = \frac{u(x)}{v(x)},\ \ \theta(\lambda,w,0) = \gamma(\lambda). \tag{2.5}$$

Now we consider the linear Dirac problem

$$\begin{cases} (\ell w)(x) = \lambda w(x),\ x\in(0,\pi), \\ U_1(\lambda,w) = 0, \\ U_2(\lambda,w) = 0, \end{cases} \tag{2.6}$$

that obtained from (1.1)-(1.3) by setting $g\equiv 0$.

Remark 2.1. Problem (2.6) was studied in [1], where it was shown that the eigenvalues λ_k, $k \in \mathbb{Z}$, of this problem are real and simple, and they can be numbered in ascending order on the real axis

$$\ldots < \lambda_{-k} < \ldots < \lambda_{-2} < \lambda_{-1} < \lambda_0 < \lambda_1 < \lambda_2 < \ldots < \lambda_k < \ldots ,$$

so that the corresponding angular function $\theta(\lambda_k, w_k, x)$ at $x = 0$ and $x = \pi$ satisfies the conditions

$$\theta(\lambda_k, w_k, 0) = \gamma(\lambda_k), \quad \theta(\lambda_k, w_k, \pi) = \delta(\lambda_k) + \pi k, \tag{2.7}$$

where $w_k(x)$, $k \in \mathbb{Z}$, is an eigenvector-function corresponding to the kth eigenvalue λ_k.

Note that the spectral properties of problem (2.6) was studied in [12] in the case of $\alpha, \beta \in [-\pi/2, \pi/2]$.

We define the integers m_{-1} and m_1 as follows:

$$\begin{cases} m_{-1} = \max\left\{k \in \mathbb{Z} : \lambda_k + p(x) < 0,\ \lambda_k + r(x) < 0,\ x \in [0, \pi]\right\}, \\ \\ m_1 = \min\left\{k \in \mathbb{Z} : \lambda_k + p(x) > 0,\ \lambda_k + r(x) > 0,\ x \in [0, \pi]\right\}. \end{cases} \tag{2.8}$$

Remark 2.2. From (2.8) it follows that the function $\theta(\lambda_k, w_k, x)$ satisfies the first and second parts of statement (ii) of [6, Theorem 2.1] for $k \geq m_1$ and $k \leq m_{-1}$, respectively.

From now on ν will denote an element of $\{+, -\}$ that is, either $\nu = +$ or $\nu = -$.

For each $k \in \mathbb{Z}$, $k \geq m_1$ or $k \leq m_{-1}$, each $\lambda \in \mathbb{R}$ and each ν we define the set $S_{k,\lambda}^{\nu}$ of functions $w = \begin{pmatrix} u \\ v \end{pmatrix} \in S$ which satisfy the following conditions:

(i) $\theta(\lambda, w, \pi) = \delta(\lambda) + k\pi$;

(ii) if $k \geq m_1$, then for fixed (λ, w), as x increases, the function θ cannot tend to a multiple of $\pi/2$ from above, and as x decreases, the function θ cannot tend to a multiple of $\pi/2$ from below; if $k \leq m_{-1}$, then for fixed (λ, w), as x increases, the function θ cannot tend to a multiple of $\pi/2$ from below, and as x decreases, the function θ cannot tend to a multiple of $\pi/2$ from above.

(iii) the function $\nu u(x)$ is positive in a deleted neighborhood of the point $x = 0$.

Let $S_{k,\lambda} = S_{k,\lambda}^{+} \cup S_{k,\lambda}^{-}$. By Remarks 2.1 and 2.2 for each $\lambda \in \mathbb{R}$ the sets $S_{k,\lambda}^{+}$, $S_{k,\lambda}^{-}$ and $S_{k,\lambda}$, $k \geq m_1$ and $k \leq m_{-1}$, are nonempty. It follows from the definition of these sets that for each fixed $\lambda \in \mathbb{R}$ they are disjoint and open in E. Moreover, if $w = \begin{pmatrix} u \\ v \end{pmatrix} \in \partial S_{k,\lambda}^{\nu}$ for some $\lambda \in \mathbb{R}$, then there exists $\xi \in [0, \pi]$ such that $|w(\xi)| = 0$, i.e., $u(\xi) = v(\xi) = 0$ [7, Remark 2.7].

For each $k \in \mathbb{Z}$, $k \geq m_1$ or $k \leq m_{-1}$, and each ν let S_k and S_k^{ν} be the subsets of S defined by

$$S_k = \bigcup_{\lambda \in \mathbb{R}} S_{k,\lambda} \text{ and } S_k^{\nu} = \bigcup_{\lambda \in \mathbb{R}} S_{k,\lambda}^{\nu},$$

respectively. Note that the sets S_k and S_k^{ν} are disjoint and open in E. Moreover, if $w = \begin{pmatrix} u \\ v \end{pmatrix} \in \partial S_k^{\nu}$, then there exists $\xi \in [0, \pi]$ such that $u(\xi) = v(\xi) = 0$.

Let $\hat{E}$ be the Banach space $E \oplus \mathbb{R}^2$ with the norm

$$||\hat{w}||_0 = \left\| \begin{pmatrix} w \\ l \\ s \end{pmatrix} \right\|_0 = ||w|| + |l| + |s|,$$

and let

$$\hat{S} = \{\hat{w} = (w, l, s)^t \in \hat{E} : w \in S\}.$$

For each $k \in \mathbb{Z}$, $k \geq m_1$ or $k \leq m_{-1}$, and each ν let $\hat{S}_k^\nu$ the subset of $\hat{S}$ given by

$$\hat{S}_k^\nu = \left\{ \hat{w} = \begin{pmatrix} w \\ l \\ s \end{pmatrix} : w \in S_k^\nu \right\} \text{ and } \hat{S}_k = \hat{S}_k^+ \cup \hat{S}_k^-.$$

Let A is a linear operator defined in $\hat{E}$ by

$$Aw = A \begin{pmatrix} w \\ l \\ s \end{pmatrix} = \begin{pmatrix} \ell w \\ a_0 v(0) + b_0 u(0) \\ a_1 v(\pi) + b_1 u(\pi) \end{pmatrix} \tag{2.9}$$

on the domain

$$D(L) = \left\{ \hat{w} = \left(\begin{pmatrix} w \\ l \\ s \end{pmatrix} \right) \in \hat{E} : w \in C^1([0, \pi]; \mathbb{R}^2),\ l = -\left(v(0) \cos \alpha + u(0) \sin \alpha\right), \right.$$

$$\left. s = -\left(v(\pi) \cos \beta + u(\pi) \sin \beta\right) \right\}.$$

Here in the space $C^1([0, \pi] \oplus \mathbb{R}^2)$ the norm is determined by

$$||\hat{w}||_1 = \left\| \begin{pmatrix} w \\ l \\ s \end{pmatrix} \right\|_1 = ||w|| + ||w'|| + |l| + |s|.$$

It is obvious that the operator L is well defined. Then the problem (2.1) is recast in the equivalent operator form

$$A\hat{w} = \lambda \hat{w},\ \ \hat{w} \in D(A). \tag{2.10}$$

The operator A is closed in $\hat{E}$ with compact resolvent. Therefore, if $\lambda = 0$ is not an eigenvalue of the operator A, then there exists $A^{-1} : \hat{E} \to \hat{D}(A)$ which is completely continuous.

As the norms in $\mathbb{R} \times E$ and $\mathbb{R} \times \hat{E}$, we take

$$||(\lambda, w)|| = \left\{|\lambda|^2 + ||w||^2\right\}^{1/2} \text{ and } ||(\lambda, \hat{w})||_0 = \left\{|\lambda|^2 + ||\hat{w}||_0^2\right\}^{1/2},$$

respectively. By $\mathcal{B}_{\lambda, \varepsilon}$, $\varepsilon > 0$, we denote the open ball in $\mathbb{R} \times \hat{E}$ of radius ε with center at $(\lambda, \hat{0})$.

Now let the nonlinear operators $F : \mathbb{R} \times \hat{E} \to \hat{E}$ and $G : \mathbb{R} \times \hat{E} \to \hat{E}$ are defined by

$$G(\lambda, \hat{w}) = G\left(\lambda, \begin{pmatrix} w \\ l \\ s \end{pmatrix}\right) = \begin{pmatrix} g(x, w, \lambda) \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} g_1(x, w, \lambda) \\ g_2(x, w, \lambda) \\ 0 \\ 0 \end{pmatrix}. \tag{2.11}$$

Then problem (1.1)-(1.3) we can rewrite in the following equivalent operator form

$$A\hat{w} = \lambda\hat{w} + G(\lambda, \hat{w}). \tag{2.12}$$

It is obvious that in this case there is a one-to-one correspondence between the solutions of problems (1.1)-(1.3) and (2.12):

$$(\lambda, w) \leftrightarrow (\lambda, \hat{w}). \tag{2.13}$$

Let

$$\hat{A} = A^{-1} \text{ and } \hat{G} = A^{-1}G. \tag{2.14}$$

Then problem (2.10) reduces to the following equivalent form

$$\hat{w} = \lambda\hat{A}\hat{w} + \hat{G}(\lambda, \hat{w}). \tag{2.15}$$

Moreover, the linear problem (2.10) reduces to the linear problem

$$\hat{w} = \lambda\hat{A}\hat{w}. \tag{2.16}$$

Remark 2.3. The eigenvector function w_k corresponding to the eigenvalue λ_k of problem (2.16) will be unique using the requirement

$$\hat{w}_k \in \hat{S}_k^+ \text{ and } ||\hat{w}_k||_0 = 1.$$

3. Global bifurcation of solutions from infinity to problem (1.1)-(1.3)

Let $\tilde{\mathcal{C}}$ be the set of nontrivial solutions of the nonlinear Dirac problem (2.12) (also to problem (2.10)).

Remark 3.1. We add the points $\{(\lambda, \infty) : \lambda \in \mathbb{R}\}$ to our space $\mathbb{R} \times \hat{E}$ and defining an appropriate topology on the resulting set, we include that (λ, ∞) is an element of $\mathbb{R} \times \hat{E}$.

Theorem 3.1 *For each $k \in \mathbb{Z}$ and each ν there exists a component $\hat{\mathcal{C}}_k^\nu$ of the set $\hat{\mathcal{C}}$ that meet (λ_k, ∞) with respect to the set $\mathbb{R} \times \hat{S}_k^\nu$ and for this set at least one of the following statements holds:*

(i) $\hat{\mathcal{C}}_k^\nu$ meets (λ'_k, ∞) with respect to the set $\mathbb{R} \times \hat{S}_{k'}^{\nu'}$ for some $(k', \nu') \neq (k, \nu)$;

(ii) $\hat{\mathcal{C}}_k^\nu$ meets $\hat{\mathcal{R}} = \{(\lambda, \hat{0}) : \lambda \in \mathbb{R}\}$ for some $\lambda \in \mathbb{R}$;

(iii) the natural projection $P_{\hat{\mathcal{R}}}(\hat{\mathcal{C}}_k^\nu)$ onto $\hat{\mathcal{R}}$ is unbounded.

Proof. The proof will be carried out in three steps.

Step 1. Recall that the operator $\hat{A} : \hat{E} \to D(A)$ is completely continuous. Consequently, there exists positive constants C_1 such that for any $\hat{w} \in \hat{E}$ the following relation holds:

$$||\hat{A}\hat{w}||_1 \le C_1||\hat{w}||_0. \tag{3.1}$$

By (1.5) it follows from [5, Lemma 2] that for any sufficiently small $\epsilon \in (0, 1)$ there is a sufficiently large $\varrho_\epsilon > 0$ such that for any $x \in [0, \pi]$, $w \in E$, $||w|| > \varrho_\epsilon$ and $\lambda \in \Lambda$, the relation

$$|g(x, w(x), \lambda)| < \epsilon\, C_1^{-1}\, ||w|| \tag{3.2}$$

holds.

We have the following relations

$$|l| \le |u(0)| + |v(0)| \le ||w|| \text{ and } |s| \le |u(\pi)| + |v(\pi)| \le ||w||,$$

and consequently,

$$||\hat{w}||_0 \le 3||w||. \tag{3.3}$$

Let

$$\hat{\varrho}_\epsilon = 3\varrho_\epsilon \text{ and } ||\hat{w}||_0 > \hat{\varrho}_\epsilon.$$

Then, by (3.3), we have

$$||w|| > \varrho_\epsilon.$$

Hence it follows from (3.2) that for any $x \in [0, \pi]$, $\hat{w} \in E$, $||\hat{w}||_0 > \hat{\varrho}_\epsilon$ and $\lambda \in \Lambda$,

$$|g(x, w, \lambda)| < \epsilon\, C_1^{-1}\, ||w||. \tag{3.4}$$

Therefore, in view of (3.4), by (3.1) for any $\lambda \in \Lambda$ and $\hat{w} \in \hat{E}$, $||\hat{w}||_0 > \hat{\varrho}_\epsilon$, we get

$$\frac{||\hat{G}(\lambda, \hat{w})||_0}{||w||_0} = \frac{||\hat{A}(G(\lambda, \hat{w}))||_0}{||w||_0} \le C_1||G(\lambda, \hat{w})||_0 = C_1 \max_{x\in[0,\,\pi]} |g(x, w(x), \lambda)| < \epsilon, \tag{3.5}$$

which implies that

$$\hat{G}(\lambda, \hat{w}) = o(||w||_0) \text{ as } ||w||_0 \to \infty, \tag{3.6}$$

uniformly for $\lambda \in \Lambda$.

Step 2. Let $(\lambda, \hat{w}) \in \mathbb{R} \times \hat{E}$ such that $\hat{w} \ne \hat{0}$. We define the continuous operator $\hat{\mathcal{G}} : \hat{E} \to \hat{E}$ as follows:

$$\hat{\mathcal{G}}(\lambda, \hat{w}) = \begin{cases} ||\hat{w}||_0^2\, \hat{G}\left(\lambda, \frac{\hat{w}}{||w||_0^2}\right) & \text{if } \hat{w} \ne \hat{0}, \\ 0 & \text{if } \hat{w} = \hat{0}. \end{cases} \tag{3.7}$$

Let $\epsilon > 0$ be an arbitrary sufficiently small number, $\hat{\eta}_\epsilon = \frac{1}{\hat{\varrho}_\epsilon}$ and $\hat{w} \in \hat{E}$ such that $||\hat{w}||_0 < \hat{\eta}_\epsilon$. Then we have $\left\|\frac{w}{||\hat{w}||_0^2}\right\|_0 > \hat{\varrho}_\epsilon$. Hence it follows from (3.5) that

$$\left\|\hat{G}\left(\lambda, \frac{\hat{w}}{||w||_0^2}\right)\right\|_0 < \frac{\epsilon}{||w||_0} \text{ for any } \lambda \in \Lambda.$$

In view of (3.7) from the last relation for any $\lambda \in \Lambda$, $\hat{w} \in \hat{E}$, $||\hat{w}||_0 < \hat{\eta}_\epsilon$, we get

$$||\hat{\mathcal{G}}(\lambda, \hat{w})||_0 = ||\hat{w}||_0^2 \left\| \hat{G}\left(\lambda, \frac{\hat{w}}{||w||_0^2}\right) \right\|_0 < \epsilon\, ||\hat{w}||_0. \tag{3.8}$$

The relation (3.18) shows that the set

$$\hat{\mathcal{G}}(\Lambda \times \mathcal{B}_{\hat{\eta}_\epsilon}) = \left\{ \hat{\mathcal{G}}(\lambda, \hat{w}) : (\lambda, \hat{w}) \in \Lambda \times \mathcal{B}_{\hat{\eta}_\epsilon} \right\}$$

is bounded.

By (2.9) and (2.11)-(2.15) the vector-function $\bar{\hat{w}} = \begin{pmatrix} \bar{u} \\ \bar{v} \end{pmatrix} = \hat{\mathcal{G}}(\lambda, \hat{w})$ satisfies the following relation

$$B\bar{w}'(x) - P(x)\bar{w}(x) = ||\hat{w}||_0^2\, g\left(x, ||\hat{w}||_0^{-2} w(x), \lambda\right),\ x \in (0, \pi). \tag{3.9}$$

Then by (3.4) and (3.8) it follows from (3.9) that

$$|\bar{w}'(x)| \leq |p\,(x)||\bar{u}(x)| + |r(x)||\bar{v}(x)| + ||\hat{w}||_0^2\, |g(x, \tfrac{w(x)}{||\hat{w}||_0^2}, \lambda)| \leq M||\bar{w}|| + \epsilon\, C_1^{-1}\, ||w|| \leq$$

$$\epsilon\, M||\hat{w}||_0 + \epsilon\, C_1^{-1}\, ||\hat{w}||_0 < \epsilon\, (M + C_1)\, \hat{\eta}_\epsilon,$$

where $M = \max\{||p\,(x)||_\infty,\, ||r(x)||_\infty\}$. Hence it follows from Arzelà-Ascoli theorem that the set $\hat{\mathcal{G}}(\Lambda \times \mathcal{B}_{\hat{\eta}_\epsilon})$ is compact in $\mathbb{R} \times \hat{E}$, i.e. operator $\hat{\mathcal{G}}$ is completely continuous.

Step 3. Let $(\lambda, \hat{w})$ be a nontrivial solution of the problem (2.15). Setting $\hat{\mathrm{w}} = \frac{w}{||\hat{w}||_0^2}$ we have $||\hat{\mathrm{w}}||_0 = \frac{1}{||\hat{w}||_0}$ and $w = \frac{\hat{\mathrm{w}}}{||\hat{\mathrm{w}}||_0^2}$. Then dividing (2.15) by $||\hat{w}||_0^2$ and using (3.7), we get the problem

$$\hat{\mathrm{w}} = \lambda \hat{A}\hat{\mathrm{w}} + \hat{\mathcal{G}}(\lambda, \hat{\mathrm{w}}). \tag{3.10}$$

Let $\mathfrak{C} \subset \mathbb{R} \times \hat{E}$ be the set of nontrivial solutions of problem (3.10).

By (3.7) we have

$$\hat{\mathcal{G}}(\lambda, \hat{\mathrm{w}}) = o(||\hat{\mathrm{w}}||_0) \text{ as } ||\hat{\mathrm{w}}||_0 \to 0, \tag{3.11}$$

uniformly in $\lambda \in \Lambda$. Since the characteristics values of problem (2.16) are eigenvalues of problem (1.1)-(1.3) and are simple, by (3.11) it follows from [15, Theorem 1.3] that for each $k \in \mathbb{Z}$ there exists a continuum $\hat{\mathfrak{C}}_k^\nu$ of $\mathfrak{C}$ that meet $(\lambda_k, \hat{0})$ and either (i) $\hat{\mathfrak{C}}_k^\nu$ is unbounded in $\mathbb{R} \times \hat{E}$, or (ii) $\hat{\mathfrak{C}}_k^\nu$ meets $(\lambda'_k, \hat{0})$ for some $k' \neq k$. Note that we can decomposed $\hat{\mathfrak{C}}_k^\nu$ into two subcontinua $\hat{\mathfrak{C}}_k^+$ and $\hat{\mathfrak{C}}_k^-$ using the construction that is presented by E.N. Dancer in [10, pp. 1070-1071]. Consequently, there exists sufficiently small $\varepsilon_k > 0$ such that if $(\lambda, \hat{\mathrm{w}}) \in \hat{\mathfrak{C}}_k^\nu \cap \hat{\mathcal{B}}_{\lambda_k, \epsilon_k}$, then

$$\lambda = \lambda_k + o(|\varsigma|),\ \hat{\mathrm{w}} = \varsigma \hat{w}_k + \hat{\mathbf{w}} \text{ with } \hat{\mathbf{w}} = o(|\varsigma|), \tag{3.12}$$

where $\varsigma \in \mathbb{R}^\nu$ ($\mathbb{R}^+ = (0, +\infty)$ and $\mathbb{R}^+ = (-\infty, 0)$). Since for each $k \in \mathbb{Z}$, $k \geq m_1$ or $k \leq m_{-1}$, the set $\hat{S}_k^\nu$ is open in $\mathbb{R} \times \hat{E}$, by Remark 2.3, it follows from (3.12) that

$$\hat{\mathfrak{C}}_k^+ \cap \hat{\mathcal{B}}_{\lambda_k, \epsilon_k} \subset \mathbb{R} \times \hat{S}_k^+,\ \ \hat{\mathfrak{C}}_k^- \cap \hat{\mathcal{B}}_{\lambda_k, \epsilon_k} \subset \mathbb{R} \times \hat{S}_k^- \text{ for } k \geq m_1 \text{ or } k \leq m_{-1}. \tag{3.13}$$

The relations in (3.13) show that for each $k \in \mathbb{Z}$, $k \geq m_1$ or $k \leq m_{-1}$, the subcontinua $\hat{\mathfrak{C}}_k^+$ and $\hat{\mathfrak{C}}_k^-$ meet the bifurcation point $(\lambda_k, \hat{0})$ with respect to the sets $\mathbb{R} \times \hat{S}_k^+$ and $\mathbb{R} \times \hat{S}_k^-$, respectively. Moreover, by [10, Theorem 2] for each $k \in \mathbb{Z}$, $k \geq m_1$ or $k \leq m_{-1}$, and each ν the subcontinuum $\hat{\mathfrak{C}}_k^\nu$ is either unbounded in $\mathbb{R} \times \hat{E}$ or there exists some $(k', \nu') \neq (k, \nu)$ such that $\hat{\mathfrak{C}}_k^\nu$ meets $(\lambda_{k'}, \hat{0})$ with respect to the set $\mathbb{R} \times \hat{S}_{k'}^{\nu'}$. Should be noted that if $\hat{\mathfrak{C}}_k^\nu$ is unbounded in $\mathbb{R} \times \hat{E}$, then the following two cases are possible: (a) the natural projection $P_{\mathcal{R}}(\hat{\mathfrak{C}}_k^\nu)$ of $\hat{\mathfrak{C}}_k^\nu$ onto $\mathcal{R}$ is unbounded; (b) the natural projection $P_{\mathcal{R}}(\hat{\mathfrak{C}}_k^\nu)$ of $\hat{\mathfrak{C}}_k^\nu$ onto $\mathcal{R}$ is bounded.

Now we consider the following inversion (see [16, 17])

$$\mathcal{H} : (\lambda, \hat{w}) \to (\lambda, \hat{\mathrm{w}}) = \left(\lambda, \frac{\hat{w}}{||\hat{w}||_0^2}\right), \ w \in \hat{E}.$$

By (3.6) and (3.11) it follows from the above arguments that the inversion $\mathcal{H}$ turns a bifurcation at infinity problem (2.15) into a bifurcation from zero problem (3.10). In this case, the inversion $\mathcal{H}$ maps the set $\hat{\mathcal{C}}$ into $\hat{\mathfrak{C}}$ and, heuristically, interchanges points $(\lambda, \hat{0})$, $\lambda \in \mathbb{R}$, (respectively, (λ, ∞), $\lambda \in \mathbb{R}$) with points (λ, ∞), $\lambda \in \mathbb{R}$ (respectively, $(\lambda, \hat{0})$, $\lambda \in \mathbb{R}$). By $\hat{\mathcal{C}}_k^\nu$ we denote the inverse image $\mathcal{H}^{-1}(\hat{\mathfrak{C}}_k^\nu)$ of the set $\hat{\mathfrak{C}}_k^\nu$ under transformation $\mathcal{H}$. Then the statements of the theorem about the sets $\hat{\mathcal{C}}_k^\nu$ directly follow from the above-mentioned properties of the set $\hat{\mathfrak{C}}_k^\nu$ under the inversion of $\mathcal{H}$. The proof of this theorem is complete.

Now by (2.13) it follows from Theorem 3.1 the following result.

Theorem 3.1 *For each $k \in \mathbb{Z}$ and each ν there exists a component $\mathcal{C}_k^\nu$ of the set $\mathcal{C}$ that meet (λ_k, ∞) with respect to the set $\mathbb{R} \times S_k^\nu$ and for this set at least one of the following statements holds:*

(i) $\mathcal{C}_k^\nu$ meets (λ_k', ∞) with respect to the set $\mathbb{R} \times S_{k'}^{\nu'}$ for some $(k', \nu') \neq (k, \nu)$;

(ii) $\mathcal{C}_k^\nu$ meets $\mathcal{R} = \{(\lambda, \tilde{0}) : \lambda \in \mathbb{R}\}$ for some $\lambda \in \mathbb{R}$;

(iii) the natural projection $P_{\mathcal{R}}(\mathcal{C}_k^\nu)$ onto $\mathcal{R}$ is unbounded.

References

[1] Z.S. Aliyev, N.S. Aliyeva, On a spectral problem for the Dirac system with boundary conditions depending on the spectral parameter, Materials of the International Scientific Conference on "Modern Problems of Mathematics and Mechanics" dedicated to the 100th anniversary of the birth of the National Leader of the Azerbaijani People, Heydar Aliyev, Baku, April 26-28, 2023, p. 161-162.

[2] N.S. Aliyeva, Global bifurcation from zero in a nonlinear Dirac system with boundary conditions depending on a spectral parameter, Materials of the Republican Scientific Conference on "Diferensial and Integral Operators" dedicated to the 100th anniversary of the birth of the National Leader of the Azerbaijani People, Heydar Aliyev, Baku, November, 28-29, 2023, p. 80-82.

[3] Z.S. Aliyev, S.S. Hadiyeva, N.A. Ismayilova, Global bifurcation from infinity in some nonlinear Sturm-Liouville problems. *Bull. Malays. Math. Sci. Soc.* **46**, (2023), 1-14.

[4] Z.S. Aliyev, N.A. Mustafayeva, Bifurcation of solutions from infinity for certain nonlinear eigenvalue problems of fourth-order ordinary differential equations, *Electron. J. Differ. Equ.* **2018** (2018), No. 98, 1-19.

[5] Z.S. Aliyev ZS, N.A. Neymatov, H.Sh. Rzayeva, Unilateral global bifurcation from infinity in nonlinearizable one-dimensional Dirac problems. *Int. J. Bifur. Chaos.* 2021; 31(1): 1-10.

[6] Z.S. Aliyev, H.Sh. Rzayeva, Oscillation properties for the equation of the relativistic quantum theory, *Appl. Math. Comput.* **271** (2015), 308-316.

[7] Z.S. Aliyev, H.Sh. Rzayeva, Global bifurcation for nonlinear Dirac problems, *Electron. J. Qual. Theory Differ. Equ.* (46) (2016), 1-14.

[8] F.V. Atkinson, *Discrete and Continuous Boundary Problems*, Academic Press, New York, London, 1964.

[9] G. Dai, Global bifurcation from intervals for Sturm-Liouville problems which are not linearizable, *Elec. J. Qual. Theory of Diff. Equat.* (65) (2013), 1-7.

[10] E.N. Dancer, On the structure of solutions of nonlinear eigenvalue problems, *Indiana Univ. Math. J.* **23**(11) (1974), 1069–1076.

[11] D.D. Ivanenko, Notes to the theory of interaction via particles, *Zhurn. Experim. Teoret. Fiz.* **8** (1938), 260-266.

[12] N.B. Kerimov, A boundary value problem for the Dirac system with a spectral parameter in the boundary conditions, *Differ. Equ.* **38**(2) (2002), 164-174.

[13] B.M. Levitan, I.S. Sargsjan, *Introduction to Spectral theory; Selfadjoint ordinary differential operators* , Transl. Math. Monogr. 39, Amer. Math. Soc., Providence, R.I., 1975.

[14] J. Przybycin, Some applications of bifurcation theory to ordinary differential equations of the fourth order, *Ann. Polon. Math.* **53**(2) (1991), 153-160.

[15] P.H. Rabinowitz, Some global results for nonlinear eigenvalue problems, *J. Funct. Anal.* **7** (1971), 487-513.

[16] P.H. Rabinowitz, On bifurcation from infinity, *J. Differential Equations* **14**(3) (1973), 462-475.

[17] B.P. Rynne, Bifurcation from zero or infinity in Sturm-Liouville problems which are not linearizable, *J. Math. Anal. Appl.* **228**(1) (1998), 141-156.

[18] C.A. Stuart, Solutions of large norm for non-linear Sturm-Liouville problems, *Quart. J. Math.* **24**(2) (1973), 129-139.

[19] B. Thaller, *The Dirac Equation*, Springer, Berlin, 1992.

[20] J.F. Toland, Asymptotic linearity and non-linear eigenvalue problems, *Quart. J. Math.* **24**(2) (1973), 241-250.

Nigar G. Aliyeva
Institute of Mathematics and Mechanics of Ministry of Science and Education of Azerbaijan, Baku, Azerbaijan
E-mail: nigaraliyeva1205@gmail.com

Received 27 September 2023
Accepted 19 November 2023

Caspian Journal of Applied Mathematics, Ecology and Economics
V. 11, No 2, 2023, December
ISSN 1560-4055

Nodal Solutions of Some Nonlinear Dirac Problems

Humay Sh. Rzayeva

Abstract. In this paper we consider the nonlinear boundary value problem for the one-dimensional Dirac operator. We show the existence of pair of nodal solutions of this problem. The proof of our main result based on a method on bifurcation from zero and infinity.

Key Words and Phrases: nonlinear Dirac equation, eigenvalue, eigenvector-function, bifurcation from zero, bifurcation from infinity, nodal solution

2010 Mathematics Subject Classifications: 34A30, 34B15, 34C23, 34K29, 47J10, 47J15

---

1. Introduction

We consider the following nonlinear Dirac equation

$$(\ell w)(x) \equiv Bw'(x) - P(x)w(x) = sf(w(x)),\ 0 < x < \pi, \tag{1.1}$$

with the boundary conditions

$$U_1(w) := (\sin\alpha,\, \cos\alpha)\, w(0) = v(0)\cos\alpha + u\,(0)\sin\alpha\ = 0, \tag{1.2}$$

$$U_2(w) := (\sin\beta,\, \cos\beta) w(\pi) = v(\pi)\cos\beta + u\,(\pi)\sin\beta = 0, \tag{1.3}$$

where

$$B = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},\ P(x) = \begin{pmatrix} p\,(x) & 0 \\ 0 & r(x) \end{pmatrix},\ w(x) = \begin{pmatrix} u(x) \\ v(x) \end{pmatrix},$$

$\lambda \in \mathbb{C}$ is an eigenvalue parameter, $p\,(x),\, r(x) \in C([0,\pi];\mathbb{R})$, $\alpha,\, \beta$, are real constants such that $0 \le \alpha,\, \beta < \pi$. Here s is a nonzero real number and the function $f = \begin{pmatrix} f_1 \\ f_2 \end{pmatrix} \in C\left(\mathbb{R}^2 \times \mathbb{R}^2; \mathbb{R}^2\right)$ satisfies the following condition:

$$f(w) = \gamma w + h(w) \text{ and } f(w) = \delta w + \hbar(w), \tag{1.4}$$

where $\gamma,\, \delta$ $(\gamma \neq \delta)$ are positive constants and

---

$$h(w) = \begin{pmatrix} h_1(w) \\ h_2(w) \end{pmatrix} = o(|w|) \text{ as } |w| \to 0, \tag{1.5}$$

$$\hbar(w) = \begin{pmatrix} \hbar_1(w) \\ \hbar_2(w) \end{pmatrix} = o(|w|) \text{ as } |w| \to \infty \tag{1.6}$$

(here by $|\cdot|$ we denote the norm in $\mathbb{R}^2$).

The Dirac equations describing spin-1/2 particles, such as electrons and positrons, play the important role in both physics and mathematics. The nonlinear Dirac equations describes the self-action in quantum electrodynamics and used as an effective theory in atomic and gravitational physics (see [14]).

The existence of nodal solutions of boundary value problems for ordinary differential operators of second and fourth orders was study in [5-11]. In these papers, the authors, using the bifurcation technique, show the existence of solutions of problems they consider, contained in classes of functions with a fixed usual oscillation count.

In this paper, we will show the existence of nodal solutions of problem (1.1)-(1.3) using global bifurcation results from zero and infinity that we previously obtained for nonlinear Dirac eigenvalue problems.

2. Preliminary

Let $(b.c.)$ be denote the set of vector-functions that satisfies the boundary conditions (1.2) and (1.3).

We consider the following nonlinear eigenvalue problem

$$\begin{cases} (\ell w)(x) = \lambda w(x) + g(x, w(x), \lambda),\ x \in (0, \pi), \\ w \in (b.c.). \end{cases} \tag{2.1}$$

We will impose one or the other of the following conditions on the nonlinear term $g(x, w, \lambda) \in C([0, \pi] \times \mathbb{R}^2 \times \mathbb{R}; \mathbb{R}^2)$:

$$g(x, w, \lambda) = o(|w|) \text{ as } |w| \to 0, \tag{2.2}$$

or

$$g(x, w, \lambda) = o(|w|) \text{ as } |w| \to +\infty, \tag{2.3}$$

uniformly with respect to $(x, \lambda) \in [0, \pi] \times \Lambda$, for any bounded interval $\Lambda \subset \mathbb{R}$.

By conditions (2.2) and (2.3) the nonlinear problem (2.1) is linearizable both at zero and at infinity, and the corresponding linear problem is

$$\begin{cases} (\ell w)(x) = \lambda w(x),\ x \in (0, \pi), \\ w \in (b.c.). \end{cases} \tag{2.4}$$

Let E be the Banach space $C\left([0,\pi];\,\mathbb{R}^2\right)\cap\{w\,:\,U(w)=0\}$ with the norm $||w||_0 = \max\limits_{x\in[0,\,\pi]}|u(x)| + \max\limits_{x\in[0,\,\pi]}|v(x)|$. We denote by S a subset of E, which defined as follows:

$$S=\{w\in E:\ |u(x)+|v(x)|\ >0,\,\forall x\in[0,\pi]\}\,.$$

As in [2] (see also [4]), for each $w=\begin{pmatrix} u \\ v \end{pmatrix}\in S$ we define a continuous function $\theta(w,x)$ on $[0,\pi]$ by

$$\tan\theta(w,x)=\frac{v(x)}{u(x)},\ \ \theta(w,0)=-\,\alpha. \tag{2.5}$$

Theorem 2.1 [2, Theorem 3.1]. *The eigenvalues $\lambda_k, k\in\mathbb{Z}$, of the problem (2.4) are real and simple, and can be numbered in ascending order on the real axis*

$$\ldots\ <\lambda_{-k}<\ \ldots\ <\lambda_{-1}<\lambda_0<\lambda_1<\ \ldots\ <\lambda_k<\ \ldots\ ,$$

so that for the eigenvector-function $w_k(x)$, $k\in\mathbb{Z}$, corresponding to the eigenvalue λ_k, the angular function $\theta(w_k,x)$ at $x=\pi$ satisfy the condition

$$\theta(w_k,\pi)=-\beta+k\pi. \tag{2.6}$$

The eigenvector-functions $w_k(x)=\begin{pmatrix} u_k(x) \\ v_k(x) \end{pmatrix}$ have, with a suitable interpretation, the following oscillation properties: if $k>0$ and $k=0,\alpha\geq\beta$ (except the cases $\alpha=\beta=0$ and $\alpha=\beta=\pi/2$), then

$$\begin{pmatrix} s(u_k) \\ s(v_k) \end{pmatrix}=\begin{pmatrix} k-1+\chi(\alpha-\pi/2)+\chi(\pi/2-\beta) \\ k-1+\operatorname{sgn}\alpha \end{pmatrix}, \tag{2.7}$$

and if $k<0$ and $k=0,\ \alpha<\beta$, then

$$\begin{pmatrix} s(u_k) \\ s(v_k) \end{pmatrix}=\begin{pmatrix} |\,k|-1+\chi(\pi/2-\alpha)+\chi(\beta-\pi/2) \\ |\,k|-1+\operatorname{sgn}\beta \end{pmatrix}, \tag{2.8}$$

where $s(g)$ the number of zeros of the function $g\in C([0,\pi]\,;\,\mathbb{R})$ in the interval $(0,\pi)$ and

$$\chi\,(x)=\begin{cases} 0,\ \ if\ \ x\leq 0, \\ 1,\ \ if\ \ x>0. \end{cases}$$

Moreover, the functions $u_k(x)$ and $v_k(x)$ have only nodal zeros in the interval $(0,\pi)$.

For each $k\in\mathbb{Z}$ and each ν, let S_k^ν denote the set of vector functions $w\in S$ with the following properties:

(i) $\theta(w,\pi)=-\beta+k\pi$;

(ii) if $k>0$ or $k=0,\alpha\geq\beta$ (except the cases $\alpha=\beta=0$ and $\alpha=\beta=\pi/2$), then for fixed w, as x increases from 0 to π, the function θ cannot tend to a multiple of $\pi/2$ from above, and as x decreases, the function θ cannot tend to a multiple of $\pi/2$ from below; if

$k < 0$ or $k = 0, \alpha < \beta$, then for fixed w, as x increases, the function θ cannot tend to a multiple of $\pi/2$ from below, and as x decreases, the function θ cannot tend to a multiple of $\pi/2$ from above;

(iii) the function $\nu u(x)$ is positive in a deleted neighborhood of $x = 0$.

Remark 2.1. By (2.5), Theorem 2.1 and [2, Remark 2.1] we have $w_k \in S_k = S_k^- \cup S_k^+$, in other words the sets S_k^-, S_k^+ and S_k are nonempty. From the definition of the sets S_k^-, S_k^+ and S_k it can be seen that they are disjoint and open in E.

Now for problem (1.1)-(1.3) we can give global bifurcation results obtained earlier in [1, 3].

Theorem 2.2 [3, Theorem 3.1]. *Let condition (2.2) holds. Then for each $k \in \mathbb{Z}$ and each ν, there exists a continuum C_k^ν of solutions of problem (2.1) which contains $(\lambda_k, \tilde{0})$, is lies in $(\mathbb{R} \times S_k^\nu) \cup \{(\lambda_k, \tilde{0})\}$ and is unbounded in $\mathbb{R} \times E$.*

Theorem 2.3 [1, Theorem 1]. *Let condition (2.3) holds. Then for each $k \in \mathbb{Z}$ and each ν there exists a continua D_k^ν of solutions of problem (2.1) which contains (λ_k, ∞) and has the following properties:*

(i) there exists a neighborhood V_k of (λ_k, ∞) in $\mathbb{R} \times E$ such that $V_k \cap D_k^\nu \subset \mathbb{R} \times S_k^\nu$;

(ii) either D_k^ν meets (λ'_k, ∞) with respect to the set $\mathbb{R} \times S_{k'}^{\nu'}$ for some $(k', \nu') \neq (k, \nu)$, or D_k^ν meets $(\lambda, \tilde{0})$ for some $\lambda \in \mathbb{R}$, or D_k^ν has an unbounded projection onto $\mathbb{R} \times \{\tilde{0}\}$.

3. Existence of nodal solutions to the nonlinear problem (1.1)-(1.3)

In this section, for each $k \in \mathbb{Z}$ and each ν, an interval is determined for s in which there are solutions to the problem (1.1)-(1.3) contained in the set S_k^ν.

Theorem 1. *Let $\lambda_k \neq 0$ and*

$$\frac{\lambda_k}{\gamma} < s < \frac{\lambda_k}{\delta} \text{ or } \frac{\lambda_k}{\delta} < s < \frac{\lambda_k}{\gamma}$$

for some $k \in \mathbb{Z}$. Then problem has two solutions $w_{k,+}$ and $w_{k,-}$ such that $w_{k,+} \in S_k^+$ and $w_{k,-} \in S_k^-$.

Proof. By the first relation of (1.5) we can rewrite (1.1)-(1.3) it following form

$$\begin{cases} (\ell w)(x) = s\gamma w(x) + sh(w(x)),\ x \in (0, \pi), \\ w \in (b.c.). \end{cases} \tag{3.2}$$

Alongside problem (3.2) we shall consider the following nonlinear eigenvalue problem

$$\begin{cases} (\ell w)(x) = \lambda s\gamma w(x) + sh(w(x)),\ x \in (0, \pi), \\ w \in (b.c.), \end{cases} \tag{3.3}$$

or, equivalently

$$\begin{cases} (\tilde{\ell}w)(x) = \lambda w(x) + \tilde{g}(x,w),\ x \in (0,\pi), \\ w \in (b.c.), \end{cases} \tag{3.4}$$

where

$$\tilde{\ell}w = \frac{1}{s\gamma}\ell w \ \text{ and } \ \tilde{g}(x,w) = \frac{1}{\gamma}\, h(w(x)).$$

In view of condition (1.5) we have

$$\tilde{g}(x,w) = o(|w|) \ \text{ as } \ |w| \to 0, \tag{3.5}$$

uniformly with respect to $x \in [0,\pi]$. Then, in view of (3.5), by Theorem 2.2 for each $k \in \mathbb{Z}$ and each ν, there exists a continuum $\tilde{C}_k^\nu$ of solutions of problem (3.4) (also problem (3.3)) which contains $\left(\tilde{\lambda}_k, \tilde{0}\right)$, is lies in $(\mathbb{R} \times S_k^\nu) \cup \left\{(\tilde{\lambda}_k, \tilde{0})\right\}$ and is unbounded in $\mathbb{R} \times E$, where $\tilde{\lambda}_k$ is a kth eigenvalue of the linear problem

$$\begin{cases} (\tilde{\ell}w)(x) = \lambda w(x),\ x \in (0,\pi), \\ w \in (b.c.), \end{cases}$$

also of the linear problem

$$\begin{cases} (\ell w)(x) = \lambda s\gamma\, w(x),\ x \in (0,\pi), \\ w \in (b.c.). \end{cases} \tag{3.6}$$

In view of (3.6) we get

$$\tilde{\lambda}_k = \frac{\lambda_k}{s\gamma}.$$

By the second representation of (1.4) problem (3.3) will take the form

$$\begin{cases} (\ell w)(x) - s(\delta - \gamma)w(x) = \lambda s\gamma w(x) + s\hbar(w(x)),\ x \in (0,\pi), \\ w \in (b.c.), \end{cases} \tag{3.6}$$

or, equivalently

$$\begin{cases} (\hat{\ell}w)(x) = \lambda w(x) + \hat{g}(x,w),\ x \in (0,\pi), \\ w \in (b.c.), \end{cases} \tag{3.7}$$

where

$$\hat{\ell}w = \frac{1}{s\gamma}\ell w - \left(\frac{\delta}{\gamma} - 1\right) w \ \text{ and } \ \hat{g}(x,w) = \frac{1}{\gamma}\, h(w(x)).$$

In view of condition (1.6) for any bounded intervals $\Lambda \subset \mathbb{R}$

$$\hat{g}(x,w,\lambda) = o(|w|) \ \text{ as } \ |w| \to +\infty, \tag{3.8}$$

uniformly with respect to $(x,\lambda) \in [0,\pi] \times \Lambda$. Then, by (3.8), it follows from Theorem 2.3 that for each $k \in \mathbb{Z}$ and each ν there exists a continua $\hat{C}_k^\nu$ of solutions of problem (3.7) (also of problems (3.7) and (3.2)) which contains $(\hat{\lambda}_k, \infty)$ and has the following properties:

(i) there exists a neighborhood $\hat{V}_k$ of $(\hat{\lambda}_k, \infty)$ in $\mathbb{R} \times E$ such that $\hat{V}_k \cap \hat{C}_k^\nu \subset \mathbb{R} \times S_k^\nu$;

(ii) either $\hat{C}_k^\nu$ meets $(\hat{\lambda}'_k, \infty)$ with respect to the set $\mathbb{R} \times S_{k'}^{\nu'}$ for some $(k', \nu') \neq (k, \nu)$, or $\hat{C}_k^\nu$ meets $(\lambda, \tilde{0})$ for some $\lambda \in \mathbb{R}$, or $\hat{C}_k^\nu$ has an unbounded projection onto $\mathbb{R} \times \{\tilde{0}\}$, where $\hat{\lambda}_k$ is a kth eigenvalue of the linear problem

$$\begin{cases} (\hat{\ell} w)(x) = \lambda w(x), \ x \in (0, \pi), \\ w \in (b.c.), \end{cases}$$

also of the linear problem

$$\begin{cases} (\ell w)(x) = s(\lambda\gamma + \delta - \gamma) w(x) + s\hbar(w(x)), \ x \in (0, \pi), \\ w \in (b.c.). \end{cases} \tag{3.9}$$

By (3.9) we get

$$\hat{\lambda}_k = \frac{\lambda_k}{s\gamma} - \left(\frac{\delta}{\gamma} - 1\right).$$

Since conditions (1.5) and (1.6) are satisfied simultaneously, it follows from the proof of the first part of [13, Theorem 3.3] that $\hat{C}_k^\nu \subset \mathbb{R} \times S_k^\nu$, and consequently, the first alternative of property (ii) of $\hat{C}_k^\nu$ cannot hold, i.e., $\hat{C}_k^\nu$ cannot meets (λ'_k, ∞) for any $k' \neq k$. Then either $\hat{C}_k^\nu$ meets $(\lambda, 0)$ for some $\lambda \in \mathbb{R}$, or the projection $\hat{C}_k^\nu$ onto $\mathbb{R} \times \{\tilde{0}\}$ is unbounded.

Let $\hat{C}_k^\nu$ meets $(\lambda, \tilde{0})$ for some $\lambda \in \mathbb{R}$. Then, using [12, Lemma 1.24], we can show that $\lambda = \tilde{\lambda}_k$. Moreover, if $\tilde{C}_k^\nu$ meets (λ, ∞) for some $\lambda \in \mathbb{R}$, then from property (i) of $\hat{C}_k^\nu$ it follows that $\lambda = \hat{\lambda}_k$.

If the projection $\hat{C}_k^\nu$ onto $\mathbb{R} \times \{\tilde{0}\}$ is unbounded, then there exists $\{(\mu_n, z_n)\}_{n=1}^\infty \subset (\hat{C}_k^\nu \backslash \{(\hat{\lambda}_k, \infty)\}) \subset \mathbb{R} \times S_k^\nu$ $\left(z_n = \begin{pmatrix} z_{1n} \\ z_{2n} \end{pmatrix}\right)$ such that either

$$\lim_{n \to \infty} \mu_n \to +\infty \text{ or } \lim_{n \to \infty} \mu_n \to -\infty.$$

We introduce the notations:

$$\begin{aligned} \varphi_{n,1}(x) &= \tfrac{\hbar_1(z_n(x)) z_{n,1}(x)}{z_{n,1}^2(x) + z_{n,2}^2(x)}, \ \psi_{n,1}(x) = \tfrac{\hbar_1(z_n(x)) z_{n,2}(x)}{z_{n,2}^2(x) + z_{n,2}^2(x)} \\ \varphi_{n,2}(x) &= \tfrac{\hbar_2(z_n(x)) z_{n,1}(x)}{z_{n,1}^2(x) + z_{n,2}^2(x)}, \ \psi_{n,2}(x) = \tfrac{\hbar_2(z_n(x)) z_{n,1}(x)}{z_{n,1}^2(x) + z_{n,2}^2(x)}. \end{aligned} \tag{3.10}$$

Let $\epsilon_0 > 0$ be a fixed small number. Then, by conditions (1.5) and (1.6), there exist a small $\rho_0 > 0$ and a large $\varrho_0 > 0$ such that

$$\frac{|h(w)|}{|w|} < \epsilon_0 \text{ for any } w \in S, \ |w| < \rho_0, \tag{3.11}$$

$$\frac{|\hbar(w)|}{|w|} < \epsilon_0 \text{ for any } w \in S, \ |w| > \varrho_0. \tag{3.12}$$

Since $\hbar \in C\left(\mathbb{R}^2 \times \mathbb{R}^2; \mathbb{R}^2\right)$ there exists $K_0 > 0$ such that

$$\frac{|\hbar(w)|}{|w|} < K_0 \text{ for any } w \in E,\ \rho_0 \leq |w| \leq \varrho_0. \tag{3.13}$$

By (1.4) and (3.11) for any $w \in S$ such that $|w| < \rho_0$ we get

$$|\hbar(w)| = |(\gamma - \delta)w + h(w)| \leq (|\gamma - \delta| + \epsilon_0)|w|. \tag{3.14}$$

Thus, it follows from (3.12)-(3.14) that

$$|\hbar(w)| \leq M|w|,\ w \in S, \tag{3.15}$$

where

$$M = \max\{K_0, |\gamma - \delta| + \epsilon_0\}.$$

In view of (3.15), by (3.10) we get

$$\begin{array}{ll} |\varphi_{n,1}(x)| \leq \frac{|\hbar_1(z_n(x))|\,|z_{n,1}(x)|}{z_{n,1}^2(x)+z_{n,2}^2(x)} \leq 2M, & |\psi_{n,1}(x)| \leq \frac{|\hbar_1(z_n(x))|\,|z_{n,2}(x)|}{z_{n,1}^2(x)+z_{n,2}^2(x)} \leq 2M, \\ |\varphi_{n,2}(x)| \leq \frac{|\hbar_2(z_n(x))|\,|z_{n,1}(x)|}{z_{n,1}^2(x)+z_{n,2}^2(x)} \leq 2M, & |\psi_{n,2}(x)| \leq \frac{|\hbar_2(z_n(x))|\,|z_{n,2}(x)|}{z_{n,1}^2(x)+z_{,2}^2(x)} \leq 2M. \end{array} \tag{3.16}$$

By (3.10) it follows from (3.9) that (μ_n, z_n), $n \in \mathbb{N}$, solves the linear problem

$$\left\{ \begin{array}{l} (\ell w)(x) - sP_n(x)w(x) = s(\lambda\gamma + \delta - \gamma)w(x),\ x \in (0, \pi), \\ w \in (b.c.), \end{array} \right. \tag{3.17}$$

where

$$P_n(x) = \left(\begin{array}{cc} \varphi_{n,1}(x) & \psi_{n,1}(x) \\ \varphi_{n,2}(x) & \psi_{n,2}(x) \end{array} \right).$$

By [3, theorem 2.4] we have

$$\begin{gathered} \theta'(z_n, x) = \\ s(\mu_n\gamma + \delta - \gamma) + p(x)\cos^2\theta(z_n, x) + r(x)\sin^2\theta(z_n, x) + s\varphi_{n,1}(x)\cos^2\theta(z_n, x) + \\ s\psi_{n,2}(x)\sin^2\theta(z_n, x) + \tfrac{1}{2}s\left\{\psi_{n,1}(x) + \varphi_{n,2}(x)\right\}\sin 2\theta(z_n, x). \end{gathered} \tag{3.18}$$

Integrating (3.18) from 0 to π and using (2.5), (2.6) we get

$$\begin{gathered} s(\mu_n\gamma + \delta - \gamma) = k\pi + \alpha - \beta - \int_0^\pi \left\{p\,(x)\cos^2\theta(z_n, x) + r(x)\sin^2\theta(z_n, x)\right\} dx - \\ \left\{ \int_0^\pi \left\{\delta + \varphi_{n,1}(x)\cos^2\theta(z_n, x) + \psi_{n,2}(x)\sin^2\theta(z_n, x)\right\} dx + \right. \\ \left. \tfrac{1}{2}\int_0^\pi \left\{\psi_{n,1}(x) + \varphi_{n,2}(x)\right\}\sin 2\theta(z_n, x) dx \right\}. \end{gathered} \tag{3.19}$$

The left side of (3.19) for sufficiently large n takes on sufficiently large values in absolute value, while by p, $r \in C([0,\pi];\mathbb{R})$ and (3.16) the right side remains bounded, which gives a contradiction. Therefore, the projection of $\hat{C}_k^{\nu}$ onto $\mathbb{R}\times\{\tilde{0}\}$ is bounded, and consequently, $\hat{C}_k^{\nu}$ meets $(\tilde{\lambda}_k, 0)$. Similarly, we can show that the projection of $\tilde{C}_k^{\nu}$ onto $\mathbb{R}\times\{\tilde{0}\}$ is also bounded, which implies that $\tilde{C}_k^{\nu}$ meets $(\hat{\lambda}_k, \infty)$. Therefore, for $k \in \mathbb{Z}$ and each ν we have

$$\tilde{C}_k^{\nu} = \hat{C}_k^{\nu}. \tag{3.20}$$

From (3.2) it is clear that the solution to this problem (λ, w) with $\lambda = 1$ is also solution to problem (1.1)-(1.3). Since $\tilde{C}_k^{\nu}$ is connected, it follows that for the existence of a solution $w \in E$ of problem (1.1)-(1.3) contained in S_k^{ν} for some $k \in \mathbb{Z}$, by (3.20), it is sufficient that on the real axis $\mathbb{R}$ the point $\frac{\lambda_k}{s\gamma}$ lies to the left of 1 and the point $\frac{\lambda_k}{s\gamma} - \left(\frac{\delta}{\gamma} - 1\right)$ lies to the right of 1, or the point $\frac{\lambda_k}{s\gamma} - \left(\frac{\delta}{\gamma} - 1\right)$ lies to the right of 1, and the point $\frac{\lambda_k}{s\gamma}$ lies to the left of 1.

If $\lambda_k > 0$, $s > 0$ and $\frac{\lambda_k}{\gamma} < s < \frac{\lambda_k}{\delta}$, then we have

$$\frac{\lambda_k}{s\gamma} < 1 \text{ and } 1 < \frac{\lambda_k}{s\delta}.$$

By conditions $s > 0$, $\delta > 0$ and $\gamma > 0$ we get

$$1 < \tfrac{\lambda_k}{s\delta} \Longrightarrow s\delta < \lambda_k \Longrightarrow s\tfrac{\delta}{\gamma} < \tfrac{\lambda_k}{\gamma} \Longrightarrow s\left(1 + \tfrac{\delta}{\gamma} - 1\right) < \tfrac{\lambda_k}{\gamma} \Longrightarrow$$

$$s < \tfrac{\lambda_k}{\gamma} - s\left(\tfrac{\delta}{\gamma} - 1\right) \Longrightarrow 1 < \tfrac{\lambda_k}{s\gamma} - \left(\tfrac{\delta}{\gamma} - 1\right).$$

Therefore, we have the following relation

$$\frac{\lambda_k}{s\gamma} < 1 < \frac{\lambda_k}{s\gamma} - \left(\frac{\delta}{\gamma} - 1\right).$$

Other cases are considered similarly. The proof of this theorem is complete.

References

[1] Z.S. Aliyev ZS, N.A. Neymatov, H.Sh. Rzayeva, Unilateral global bifurcation from infinity in nonlinearizable one-dimensional Dirac problems. *Int. J. Bifur. Chaos.* 2021; 31(1): 1-10.

[2] Z.S. Aliyev, H.Sh. Rzayeva, Oscillation properties for the equation of the relativistic quantum theory, *Appl. Math. Comput.* **271** (2015), 308-316.

[3] Z.S. Aliyev, H.Sh. Rzayeva, Global bifurcation for nonlinear Dirac problems, *Electron. J. Qual. Theory Differ. Equ.* (46) (2016), 1-14.

[4] F.V. Atkinson, *Discrete and Continuous Boundary Problems*, Academic Press, New York, London, 1964.

[5] G. Dai, Global bifurcation from intervals for Sturm-Liouville problems which are not linearizable, *Elec. J. Qual. Theory of Diff. Equat.* (65) (2013), 1-7.

[6] L.H. Erbe, H.Y. Wang, On the existence of positive solutions of ordinary differential equations, *Proc. Amer. Math. Soc.* **120**(3) (1994), 743–748.

[7] J. Henderson, H.Y. Wang, Positive solutions for nonlinear eigenvalu eproblems, J.Math.Anal.Appl. **208** (1997), 252-259.

[8] R. Ma and G. Dai, Global bifurcation and nodal solutions for a Sturm-Liouville problem with a nonsmooth nonlinearity, *J. Funct. Anal.* **265** (2013), 1443-1459.

[9] R. Ma, B.Thompson, Nodal solutions for nonlinear eigenvalue problems, *Nonlinear Anal.* **59** (2004), 707–718.

[10] R. Ma, B. Thompson, Multiplicity results for second-order two-point boundary value problems with nonlinearities across several eigenvalues, *Appl. Math. Lett.* **18**(5) (2005), 587-595.

[11] Y. Naito, S. Tanaka, On the existence of multiple solutions of the boundary value problem for nonlinear second-order differential equations, *Nonlinear Anal.* **56**(4) (2004), 919-935

[12] P.H. Rabinowitz, Some global results for nonlinear eigenvalue problems, *J. Funct. Anal.* **7**(3) (1971), 487-513.

[13] B.P. Rynne, Bifurcation from zero or infinity in Sturm-Liouville problems which are not linearizable, *J. Math. Anal. Appl.* **228**(1) (1998), 141-156.

[14] B. Thaller, *The Dirac Equation*, Springer, Berlin, 1992.

Humay Sh. Rzayyeva
Institute of Mathematics and Mechanics of Ministry of Science and Education of Azerbaijan, Baku, Azerbaijan
E-mail: humay_rzayeva@bk.ru

Received 01 October 2023
Accepted 21 November 2023

Caspian Journal of Applied Mathematics, Ecology and Economics
V. 11, No 2, 2023, December
ISSN 1560-4055

On basicity of eigenfunctions of one spectral problem with the discontinuity point in Morrey-Lebesgue spaces

T.B. Gasymov*, A.Q. Akhmedov, R.J. Taghiyeva

Abstract. In this paper is studied the spectral problem for a discontinuous second order differential operator with a spectral parameter in transmission conditions, that arises by solving the problem on vibrations of a loaded string with the free ends. An abstract theorem on the stability of the basis properties of multiple systems in a Banach space with respect to certain transformations is proved. This fact is used in the proof of theorems on the basicity of eigenfunctions of a discontinuous differential operator in Morrey type spaces.

Key Words and Phrases: spectral problem, eigenfunctions, basicity, Morrey spaces.

2010 Mathematics Subject Classifications: 34B24

---

1. Introduction

We consider a model eigenvalue problem for the discontinuous second order differential operator

$$-y''(x) = \lambda y(x), \ x \in \left(0, \frac{1}{3}\right) \bigcup \left(\frac{1}{3}, 1\right) \tag{1}$$

with boundary conditions

$$y'(0) = y'(1) = 0 \tag{2}$$

and with the following discontinuity conditions

$$\left.\begin{array}{l} y(\frac{1}{3}-0) = y(\frac{1}{3}+0), \\ y'(\frac{1}{3}-0) - y'(\frac{1}{3}+0) = \lambda m y(\frac{1}{3}), \end{array}\right\} \tag{3}$$

where λ is the spectral parameter, m is a non-zero complex number. Such spectral problems arise when the problem of vibrations of a loaded string with free ends is solved by applying the Fourier method [1-3]. The spectral problems with a discontinuity conditions inside the interval play an important role in mathematics, mechanics, physics and other fields of science. The applications of boundary value problems with discontinuity conditions inside the interval are connected with discontinuous material properties. Nowadays

*Corresponding author.

there is a number of papers dedicated to spectral problems for the ordinary differential operator with eigenparameter dependent boundary conditions. There are many papers concerning problems with discontinuity conditions. One can find the similar works in [4-7].

One of the commonly used methods for solving partial differential equations is the method of separation of variables. This method yields the appropriate spectral problem and in order to justify the method, it is very important the question of the expansion of functions of certain class on eigen and association functions of the discrete differential operators. The study of spectral properties of some discrete differential operators motivates the development of new methods for constructing basis. In this context, much attention has been given to the study of basis properties (completeness, minimality and basicity) of systems of special functions, which are frequently eigen and associated functions of differential operators. Additionally, various methods for examining these properties were proposed. Example, one of the works is [8]. Then various perturbation methods are applied. This direction has been well developed (see [9]). In the case of discontinuous differential operators, there appear systems of eigenfunctions whose basicity cannot be investigated by previously known methods. In the work [10,11] is considered an abstract approach to the problem described above and is proposed a new method for constructing bases, which has wide applications in the spectral theory of discontinuous differential operators.

In the paper [12] the problem of oscillation of a loaded string is investigated in the case when the load is placed in the middle of the string and it is shown that an abstract method proposed in [10, 11] can be used in non-standard spaces such as a Morrey type space. The concept of Morrey space was introduced by C. Morrey in 1938. Since then, various problems related to this space have been intensively studied. Playing an important role in the qualitative theory of elliptic differential equations (see, for example, [13,14]), this space also provides a large class of examples of mild solutions to the Navier-Stokes system [15]. In the context of fluid dynamics, Morrey spaces have been used to model fluid flow when vorticity is a singular measure supported on certain sets in R^n [16]. More details about Morrey spaces can be found in [17,18]. Morrey type space is introduced in [17] (see e.g. [18]). In [19] the basicity of the exponential system, and in [20-23] - the perturbed exponential system in Morrey type spaces are proved. The present paper is continuity of [24] and [25].

2. Necessary information and preliminary results

For obtaining the main results we need some notions and facts from the theory of basis in a Banach space.

Definition 2.1. *Let X- be a Banach space. If there exists a sequence of indexes, such that $\{n_k\} \subset N$, $n_k < n_{k+1}, n_0 = 0$, and any vector $x \in X$ is uniquely represented in the* *form*

$$x = \sum_{k=0}^{\infty} \sum_{i=n_k+1}^{n_{k+1}} c_i u_i.$$

then the system $\{u_n\}_{n\in N} \in X$ *is called a basis with parentheses in* X .

For $n_k = k$ the system $\{u_n\}_{n\in N}$ forms a usual basis for X.

We need the following easily proved statements.

Statement 2.1. *Let the system* $\{u_n\}_{n\in N}$ *forms a basis with parentheses for a Banach space* X. *If the sequence* $\{n_{k+1} - n_k\}_{k\in N}$ *is bounded and the condition*

$$\sup_n \|u_n\| \, \|\vartheta_n\| < \infty,$$

holds, where $\{\vartheta_n\}_{n\in N}$*- is a biorthogonal system, then the system* $\{u_n\}_{n\in N}$ *forms a usual basis for* X.

Definition 2.2. *The basis* $\{u_n\}_{n\in N}$ *of Banach space* X *is called a p-basis, if for any* $x\in X$ *the condition*

$$\left(\sum_{n=1}^{\infty} |\langle x, \vartheta_n \rangle|^p \right)^{\frac{1}{p}} \leq M \|x\|,$$

holds, where $\{\vartheta_n\}_{n\in N}$ *- is a biorthogonal system to* $\{u_n\}_{n\in N}$.

Definition 2.3. *The sequences* $\{u_n\}_{n\in N}$ *and* $\{\phi_n\}_{n\in N}$ *of Banach space* X *are called a p- close, if the following condition holds:*

$$\sum_{n=1}^{\infty} \|u_n - \phi_n\|^p < \infty.$$

We will also use the following results from [8] (see also [24]).

Theorem 2.1. *Let* $\{x_n\}_{n\in N}$ *forms a q-basis for a Banach space* X, *and the system* $\{y_n\}_{n\in N}$ *is p-close to* $\{x_n\}_{n\in N}$, *where* $\frac{1}{p} + \frac{1}{q} = 1$.*Then the following properties are equivalent:*

a) $\{y_n\}_{n\in N}$*- is complete in* X;

b) $\{y_n\}_{n\in N}$*- is minimal in* X;

c) $\{y_n\}_{n\in N}$*- forms an isomorphic basis to* $\{x_n\}_{n\in N}$ *for* X.

Let $X_1 = X \oplus C^m$ and $\{\hat{u}_n\}_{n\in N} \subset X_1$ be some minimal system and $\left\{\hat{\vartheta}_n\right\}_{n\in N} \subset X_1^* = X^* \oplus C^m$ be its biorthogonal system:

$$\hat{u}_n = (u_n; \alpha_{n1}, ..., \alpha_{nm}) \, ; \hat{\vartheta}_n = (\vartheta_n; \beta_{n1}, ..., \beta_{nm}) \, .$$

Let $J = \{n_1, ..., n_m\}$ some set of m natural numbers. Suppose

$$\delta = \det \|\beta_{n_i j}\|_{\, i,j=\overline{1,m}}.$$

In [27] (see also [28]) has been proved the following theorem :

Theorem 2.2. *Let the system $\{\hat{u}_n\}_{n\in N}$ forms a basis for X_1.In order to the system $\{u_n\}_{n\in N_J}$, where $N_J = N\backslash J$ forms a basis for X it is necessary and sufficient that the condition $\delta \neq 0$ be satisfied. In this case the biorthogonal system to $\{u_n\}_{n\in N_J}$ is defined by*

$$\vartheta_n^* = \frac{1}{\delta}\begin{vmatrix} \vartheta_n & \vartheta_{n1} & \dots & \vartheta_{nm} \\ \beta_{n1} & \beta_{n_1 1} & \dots & \beta_{n_m 1} \\ \dots & \dots & \dots & \dots \\ \beta_{nm} & \beta_{n_1 m} & \dots & \beta_{n_m m} \end{vmatrix}.$$

In particular, if X is a Hilbert space and the system $\{u_n\}_{n\in N}$- forms a Riesz basis for X_1, then under the condition $\delta \neq 0$, the system $\{u_n\}_{n\in N_J}$ also forms a Riesz basis for X. For $\delta = 0$ the system $\{u_n\}_{n\in N_J}$ is not complete and is not minimal in X.

Let X be a Banach space and the system $\{u_{kn}\}_{k=\overline{1,m};n\in N}$ is any system in X. Let $a_{ik}^{(n)}, i,k = \overline{1,m}, n \in N$, any complex numbers. Let,

$$A_n = \left(a_{ik}^{(n)}\right)_{i,k=\overline{1,m}} \quad and \;\; \Delta_n = det A_n \;, n \in N.$$

Consider the following system in space X :

$$\hat{u}_{kn} = \sum_{i=1}^{m} a_{ik}^{(n)} u_{in}, k = \overline{1,m}; n \in N. \tag{4}$$

Following theorems have been proved in [8] (also [11])

Theorem 2.3. *If the system $\{u_{kn}\}_{k=\overline{1,m};n\in N}$ forms basis for X and*

$$\Delta_n \neq 0, \forall n \in N \tag{5}$$

then the system $\{\hat{u}_{kn}\}_{k=\overline{1,m};n\in N}$ forms basis with parentheses for X. If in addition the following conditions

$$\sup_n \left\{\|A_n\|, \left\|A_n^{-1}\right\|\right\} < \infty, \qquad \sup_n \left\{\|u_{kn}\|, \|\vartheta_{kn}\|\right\} < \infty, \tag{6}$$

hold, where $\{\vartheta_{kn}\}_{k=\overline{1,m};n\in N} \subset X^*$- is biorthogonal to $\{u_{kn}\}_{k=\overline{1,m};n\in N}$, then the system $\{\hat{u}_{kn}\}_{k=\overline{1,m};n\in N}$ forms a usual basis for X.

We will also need some facts about the theory of Morrey-type spaces. Let Γ be some rectifiable Jordan curve on the complex plane C. By $|M|_\Gamma$ we denote the linear Lebesgue measure of the set $M \subset \Gamma$. By the Morrey-Lebesgue space $L^{p,\alpha}(\Gamma)$, $0 \leq \alpha \leq 1$, $p \geq 1$, we mean a normed space of all functions $f(\cdot)$ measurable on Γ equipped with a finite norm $\|f\|_{L^{p,\alpha}(\Gamma)}$:

$$\|f\|_{L^{p,\alpha}(\Gamma)} = \sup_B \left(\left|B\bigcap\Gamma\right|_\Gamma^{\alpha-1} \int_{B\bigcap\Gamma} |f(\xi)|^p \, |d\xi|\right)^{\frac{1}{p}} < +\infty.$$

$L^{p,\alpha}(\Gamma)$ is a Banach space and $L^{p,1}(\Gamma) = L_p(\Gamma)$, $L^{p,0}(\Gamma) = L_\infty(\Gamma)$. The embedding $L^{p,\alpha_1}(\Gamma) \subset L^{p,\alpha_2}(\Gamma)$ is valid for $0 \leq \alpha_1 \leq \alpha_2 \leq 1$. Thus $L^{p,\alpha}(\Gamma) \subset L_p(\Gamma)$, $\forall\alpha \in [0,1]$, $\forall p \geq 1$. The case of $\Gamma \equiv [a,b]$ will be denoted by $L^{p,\alpha}(a,b)$.

Denote by $\tilde{L}^{p,\alpha}(a,b)$ linear subspace of $L^{p,\alpha}(a,b)$ consisting of functions whose shifts are continuous in $L^{p,\alpha}(a,b)$, i.e. $\|f(\cdot+\delta)-f(\cdot)\|_{L^{p,\alpha}(a,b)} \to 0$ as $\delta \to 0$.The closure of $\tilde{L}^{p,\alpha}(a,b)$ in $L^{p,\alpha}(a,b)$ will be denoted by $M^{p,\alpha}(a,b)$. In [19,21] the following theorem is proved.

Theorem 2.4. *The exponential system* $\left\{e^{i\,nt}\right\}_{n\in Z}$ *is the bases in* $M^{p,\alpha}(-\pi,\pi)$, $1<p<+\infty$, $0<\alpha\le 1$.

Using this theorem, it is easy to obtain the following

Statement 2.2. *The trigonometric systems* $\{\cos nx\}_{n=0}^{\infty}$ *forms a basis for* $M^{p,a}(0,p)$, $1<p<+\infty, 0<a\le 1$.

3. Main results

In [24] it was proved that the eigenvalues of the problem (1)-(3) are asymptotically simple and consist of $\lambda_0=0$ and two series: $\lambda_{i,n}=\rho_{i,n}^2, i=1,2; n\in Z^+$, where $Z^+=\{0\}\cup N$ and the numbers $\rho_{i,n}$ hold the following asymptotically formulas:

$$\begin{cases} \rho_{1,n}=3\pi n+\frac{3\pi}{2}+O\left(\frac{1}{n}\right) \\ \rho_{2,n}=\frac{3\pi n}{2}+\frac{3\pi}{4}+O\left(\frac{1}{n}\right). \end{cases} \tag{7}$$

Also the eigenfunctions $y_0(x)$ and $y_{i,n}(x)$ of the problem (1)-(3) corresponding to the eigenvalues $\lambda_{i,n}=(\rho_{1,n})^2, i=\overline{1,2}; n\in Z^+$ are the following form:

$$y_0(x)\equiv 1, \quad y_{i,n}(x)=\begin{cases} cos\frac{2\rho_{i,n}}{3}\ cos\rho_{i,n}\ x, x\in\left[0,\frac{1}{3}\right], \\ cos\frac{\rho_{i,n}}{3}\ cos\rho_{i,n}\ (1-x),\ \ x\in\left[\frac{1}{3},1\right], \end{cases} \quad i=1,2; n\in Z^+. \tag{8}$$

Now consider a problem on basicity of eigen and associated functions of the problem (1)-(3) in spaces $L_p(0,1)\oplus C$ and $L_p(0,1)$. Since the eigenvalues are asymptotically simple, the problem can only have a finite number of associated functions. In paper [24] we were constructed linearizing operator as following form. By $W_p^k\left(0,\frac{1}{3}\right)\oplus W_p^k\left(\frac{1}{3},1\right)$ we denoted a space functions whose contractions on segments $\left[0,\frac{1}{3}\right]$ and $\left[\frac{1}{3},1\right]$ belong correspondingly to Sobolev spaces $W_p^k\left(0,\frac{1}{3}\right)$ and $W_p^k\left(\frac{1}{3},1\right)$. Let us define the operator L in $L_p(0,1)\oplus C$ as follows:

$$D(L)=\left\{\begin{array}{l} \hat{y}\in L_p(0,1)\oplus C:\hat{y}=\left(y,my\left(\frac{1}{3}\right)\right), y\in W_p^2\left(0,\frac{1}{3}\right)\oplus W_p^2\left(\frac{1}{3},1\right), \\ y'(0)=y'(1)=0, y\left(\frac{1}{3}-0\right)=y\left(\frac{1}{3}+0\right), \end{array}\right\}$$

and for $\hat{y}\in D(L)$

$$L\hat{y}=(-y'';y'\left(\frac{1}{3}-0\right)-y'\left(\frac{1}{3}+0\right),\hat{y}\in D(L)).$$

Operator defined by these formula is a linear closed operator with dense definitional domain in $L_p(0,1)\oplus C$. Eigenvalues of the operator L and problem (1)-(3) coincide, and

$\{\hat{y}_0\} \cup \{\hat{y}_{i,n}\}_{i=1,2;\ n\in Z^+}$ are eigenvectors of the operator L , where

$$\hat{y}_0 = (1; m)\,, \qquad \hat{y}_{i,n} = \left(y_{i,n}(x)\,; my\left(\frac{1}{3}\right)\right),\ i = 1,2; n \in Z^+.$$

Theorem 3.1. *The system $\{\hat{y}_0\} \cup \{\hat{y}_{i,n}\}_{i=1,2;\ n\in Z^+}$ of eigen and associated vectors of the operator L, which linearized the problem (1)- (3) forms basis in $M^{p,a}(0,1) \oplus C, 1 < p < \infty, 0 < a \leq 1$.*

Proof. Considering (7) in (8), we will get the following functional system for the head parts of the asymptotic formulas:

$$\begin{cases} u_{1,n}(x) = \begin{cases} -\cos\left(3\pi n + \frac{3\pi}{2}\right)x\ , & x \in \left[0, \frac{1}{3}\right], \\ 0, & x \in \left[\frac{1}{3}, 1\right], \end{cases} \\ u_{2,n}(x) = \begin{cases} 0, & x \in \left[0, \frac{1}{3}\right], \\ a_n \cos\left(\frac{3\pi n}{2} + \frac{3\pi}{4}\right)(1-x) & x \in \left[\frac{1}{3}, 1\right], \end{cases} \end{cases} \tag{9}$$

where $\alpha_n = \cos\left(\frac{\pi n}{2} + \frac{\pi}{4}\right)$. If $n = 4k$ and $n = 4k+3$, $\alpha_n = \frac{1}{\sqrt{2}}$ and $n = 4k+1$ and $n = 4k+2$, $\alpha_n = -\frac{1}{\sqrt{2}}$. Then from the formulas (8) and (9) imply that, the following asymptotic relations are true:

$$\begin{cases} y_{1,n}(x) = u_{1,n}(x) + O\left(\frac{1}{n}\right) \\ y_{2,n}(x) = u_{2,n}(x) + O\left(\frac{1}{n}\right). \end{cases} \tag{10}$$

Since the operator L has compact resolvent, the system $\{\hat{y}_0\} \cup \{\hat{y}_{i,n}\}_{i=1,2;\ n\in Z^+}$ of eigenfunctions and associated vectors is minimal in $L_p(0,1) \oplus C$. The conjugate system $\{\hat{v}_0\} \cup \{\hat{v}_{i,n}\}_{i=1,2;n\in Z^+}$ is the eigenvectors and associated vectors of the conjugating operator L^* and is in the form $\hat{v}_0 = c_0(1; \overline{m})$, $\hat{v}_{i,n} = \left(v_{i,n}(x)\,; \overline{m}v_{i,n}\left(\frac{1}{3}\right)\right)$, where $v_0(x)$ and $v_{i,n}(x), i = 1,2; n \in Z^+$ are the eigen- and associated vectors of the conjugate problem and analogically we obtain the asymptotically formulas:

$$v_{i,n}(x) = \begin{cases} c_{i,n} cos\frac{2\rho_{i,n}}{3}\ cos\rho_{i,n}\ x, & x \in \left[0, \frac{1}{3}\right], \\ c_{i,n} cos\frac{\rho_{i,n}}{3}\ cos\rho_{i,n}\ (1-x)\,, & x \in \left[\frac{1}{3}, 1\right], \end{cases} \quad i = 1,2; n \in Z^+, \tag{11}$$

here $c_0, c_{i,n}$ are the normalized multipliers. We can easily calculate that, the c_{1n}, c_{2n} normalized multipliers hold

$$c_0 = \frac{1}{1+|m|^2},\quad c_{1n} = 6 + O\left(\frac{1}{n}\right), c_{2n} = 6 + O\left(\frac{1}{n}\right).$$

If we consider these formulas at (11) we will obtain for $v_{i,n}(x)$, $i = 1,2;;n \in Z^+$ the following formulas:

$$v_{1,n} = \begin{cases} -6cos\left(3\pi n + \frac{3\pi}{2}\right)x\ + O\left(\frac{1}{n}\right), & x \in \left[0, \frac{1}{3}\right] \\ O\left(\frac{1}{n}\right), & x \in \left[\frac{1}{3}, 1\right] \end{cases} \tag{12}$$

$$v_{2.n} = \begin{cases} O\left(\frac{1}{n}\right), & x \in \left[0, \frac{1}{3}\right] \\ 6a_n\cos\left(\frac{3\pi n}{2} + \frac{3\pi}{4}\right)(1-x)\ + O\left(\frac{1}{n}\right)\,, & x \in \left[\frac{1}{3}, 1\right]. \end{cases} \tag{13}$$

One can easily seen that the system (10) implies from the following system by the conversion

$$u_{i,n} = a_{i,1}e_{1,n} + a_{i,2}e_{2,n}, i = 1,2; \quad n \in Z^{+}$$

$$\begin{cases} e_{1,n}(x) = \begin{cases} \cos\left(3\pi n + \frac{3\pi}{2}\right)x, & x \in \left[0, \frac{1}{3}\right], \\ 0, & x \in \left[\frac{1}{3}, 1\right], \end{cases} \\ e_{2,n}(x) = \begin{cases} 0, & x \in \left[0, \frac{1}{3}\right], \\ \cos\left(\frac{3\pi n}{2} + \frac{3\pi}{4}\right)(1-x), & x \in \left[\frac{1}{3}, 1\right] \end{cases} \end{cases} \tag{14}$$

where the numbers $a_{i,1}$ and $a_{i,2}$, $i = 1,2$ are the elements of the following matrix

$$A = \begin{pmatrix} -1 & 0 \\ 0 & a_n \end{pmatrix}. \tag{15}$$

Note that,

$$detA = -a_n \neq 0.$$

On the other hand the system $\{e_{i,n}\}_{i=1,2,n\in Z^{+}}$ forms a basis for $M^{p,\alpha}(0,1), 1 < p < \infty$. Indeed, according to the decomposition $M^{p,\alpha}(0,1) = M^{p,\alpha}\left(0,\frac{1}{3}\right) \oplus M^{p,\alpha}\left(\frac{1}{3},1\right)$, and since the system $\{e_{1,n}\}_{n\in Z^{+}}$ forms basis in $M^{p,\alpha}\left(0,\frac{1}{3}\right)$, and the system $\{e_{1,n}\}_{n\in Z^{+}}$ forms basis in $M^{p,\alpha}\left(\frac{1}{3},1\right)$, therefore, their combination will forms a basis in $M^{p,\alpha}(0,1)$. If we take it into consideration and apply Theorem 2.3, then we obtain that the system $\{u_{i,n}\}_{i=1,2;n\in Z^{+}}$ forms basis in $M^{p,a}(0,1)$. Consider the system in $\{\hat{u}_0\} \cup \{\hat{u}_{i,n}\}_{i=1,2;n\in Z^{+}}$ in $M^{p,\alpha}(0,1) \oplus C$, where

$$\hat{u}_0 = (0,1), \quad \hat{u}_{i,n} = (u_{i,n}; \; 0), i = \overline{1,2}; \quad n \in Z^{+}. \tag{16}$$

It is clear that, the system $\{\hat{u}_0\} \cup \{\hat{u}_{i,n}\}_{i=1,2; \; n\in Z^{+}}$ forms basis in $M^{p,\alpha}(0,1) \oplus C$. Let us show that it also forms a q-basis, where $q = p/(p-1)$. One can easily check that the system $\{\hat{v}_0\} \cup \left\{\hat{\vartheta}_{i,n}\right\}_{i=1,2;n\in Z^{+}}$, which biorthogonal to it is in the following form:

$$\hat{v}_0 = \frac{1}{1+|m|^2}(1;\overline{m}), \hat{\vartheta}_{i,n} = (\vartheta_{i,n}; 0), i = 1,2; \; n \in Z^{+}, \tag{17}$$

where

$$\vartheta_{1.n} = \begin{cases} -6\cos\left(3\pi n + \frac{3\pi}{2}\right)x, & x \in \left[0, \frac{1}{3}\right] \\ 0, & x \in \left[\frac{1}{3}, 1\right] \end{cases} \tag{18}$$

$$\vartheta_{2.n} = \begin{cases} 0, & x \in \left[0, \frac{1}{3}\right] \\ 6\alpha_n\cos\left(\frac{3\pi n}{2} + \frac{3\pi}{4}\right)(1-x), & x \in \left[\frac{1}{3}, 1\right]. \end{cases} \tag{19}$$

Let $1 < p \leq 2$. Then according to inequality Hausdorf-Young for trigonometric system (see [28], p.153) for each $f \in L_p(0,1)$ the inequality

$$\left(\sum_{i=1}^{2}\sum_{n=0}^{\infty} |< f, e_{i,n} >|^q\right)^{\frac{1}{q}} \leq M\|f\|_{L_p}$$

is fulfilled, where $M > 0$ is a fixed number which does not depend on f. Taking into consideration that, the system $\{\vartheta_{i,n}\}_{i=1,2;n\in Z^+}$ implies from the system $\{e_{i,n}\}_{i=1,2,n\in Z^+}$ by conversion

$$\vartheta_{i,n} = b_{i,1}e_{1,n} + b_{i,2}e_{2,n}, i = 1, 2;\ n \in Z^+$$

where b_{i1} and $b_{i,2}$, $i = 1, 2$ are the elements of the matrix

$$B = \begin{pmatrix} -6 & 0 \\ 0 & 6\alpha_n \end{pmatrix}.$$

We obtain from here that for an arbitrary $\hat{f} \in L_p(0,1) \oplus C$ the following inequality holds:

$$\left(\left|\left\langle \hat{f}, \hat{\vartheta}_0 \right\rangle\right|^q + \sum_{i=1}^{2}\sum_{n=0}^{\infty}\left|\left\langle \hat{f}, \hat{\vartheta}_{i,n} \right\rangle\right|^q\right)^{\frac{1}{q}} \le M\left\|\hat{f}\right\|_{L_p\oplus C}.$$

Then, taking into account the embedding $M^{p,\alpha}(0,1) \subset L_p(0,1)$, we obtain that for any $\hat{f} \in M^{p,\alpha}(0,1) \oplus C$ the inequality

$$\left(\left|\left\langle \hat{f}, \hat{\vartheta}_0 \right\rangle\right|^q + \sum_{i=1}^{2}\sum_{n=0}^{\infty}\left|\left\langle \hat{f}, \hat{\vartheta}_{i,n} \right\rangle\right|^q\right)^{\frac{1}{q}} \le M\left\|\hat{f}\right\|_{M^{p,\alpha}\oplus C},$$

holds, i.e. the system $\{\hat{u}_{i,n}\}_{i=\overline{1,2};n\in N}$ forms a q-basis for $M^{p,\alpha}(0,1) \oplus C$.

Let's point

$$\hat{y}_{i,n} = \left(y_{i,n}(x); my_{i,n}\left(\frac{1}{3}\right)\right), i = 1, 2; n \in Z^+,$$

According to the formulas (8) since $y_{i,n}\left(\frac{1}{3}\right) = O\left(\frac{1}{n}\right)$, from (10) implies that the systems $\{\hat{y}_0\} \cup \{\hat{y}_{i,n}\}_{i=1,2;\ n\in Z^+}$ and $\{\hat{u}_0\} \cup \{\hat{u}_{i,n}\}_{i=1,2;\ n\in Z^+}$ are p-close,

$$\sum_{i=1}^{2}\sum_{n=0}^{\infty}\|\hat{y}_{i,n} - \hat{u}_{i,n}\|^p < \infty.$$

Thus, all the conditions of Theorem 2.1 are fulfilled and according to this theorem the system $\{\hat{y}_0\} \cup \{\hat{y}_{i,n}\}_{i=1,2;\ n\in Z^+}$ also forms an isomorphic basis to the system $\{\hat{u}_0\} \cup \{\hat{u}_{i,n}\}_{i=1,2;\ n\in Z^+}$ in $L_p(0,1) \oplus C$, therefore, it is minimal in this space and in view of the embedding

$$(L_q(0,1) \oplus C) \subset (M^{p,\alpha}(0,1) \oplus C)^*,$$

we find that it is minimal in $M^{p,\alpha}(0,1) \oplus C$. Thus, all the conditions of Theorem 2.3 hold and by this theorem, the system $\{\hat{y}_0\} \cup \{\hat{y}_{i,n}\}_{i=1,2;\ n\in Z^+}$ forms an equivalent basis to the system $\{\hat{u}_0\} \cup \{\hat{u}_{i,n}\}_{i=1,2;\ n\in Z^+}$ for space $M^{p,\alpha}(0,1) \oplus C$.

Now suppose that $p > 2$, then $1 < q < 2$. Taking into account that in this case the following inclusion is fulfilled:

$$L_p(0,1) \subset L_q(0,1), l_p \subset l_q,$$

then for $\hat{f} \in L_p(0,1) \oplus C$ we obtain:

$$\left(\left| \left\langle \hat{f}, \hat{\vartheta}_0 \right\rangle \right|^p + \sum_{i=1}^{2} \sum_{n=0}^{\infty} \left| \left\langle \hat{f}, \hat{\vartheta}_{i,n} \right\rangle \right|^p \right)^{\frac{1}{p}} \leq M \left\| \hat{f} \right\|_{L_p \oplus C} \leq M \left\| \hat{f} \right\|_{M^{p,a} \oplus C}.$$

This implies that the system $\{\hat{u}_0\} \cup \{\hat{u}_{i,n}\}_{i=1,2;\ n \in Z^+}$ forms a p- basis in $M^{p,\alpha}(0,1) \oplus C$. Besides, the systems $\{\hat{y}_0\} \cup \{\hat{y}_{i,n}\}_{i=1,2;\ n \in Z^+}$ and $\{\hat{u}_0\} \cup \{\hat{u}_{i,n}\}_{i=1,2;\ n \in Z^+}$ are q-close in $M^{p,\alpha}(0,1) \oplus C$:

$$\sum_{i=1}^{2} \sum_{n=0}^{\infty} \left\| \hat{y}_{i,n} - \hat{u}_{i,n} \right\|^q_{M^{p,\alpha} \oplus C} < \infty.$$

According to the system $\{\hat{y}_0\} \cup \{\hat{y}_{i,n}\}_{i=1,2;\ n \in Z^+}$ is minimal in $M^{p,a}(0,1) \oplus C$ and again applying the Theorem 2.1, we obtain that it is a basis in $M^{p,a}(0,1) \oplus C$ isomorphic to $\{\hat{u}_0\} \cup \{\hat{u}_{i,n}\}_{i=1,2;\ n \in Z^+}$.

Now consider the basicity $\{y_0\} \cup \{y_{i,n}\}_{i=2;n \in Z^+}^{\infty}$ of the system of eigenfunctions and associated functions of the problem (1)-(3) in $L_p(0,1)$. Applying the Theorem 2.3 , we obtain the truth of the following theorem.

Theorem 3.2. *In order the system* $\{y_{i,n}\}_{i=\overline{1,2};n \in Z^+, n \neq n_0}^{\infty}$ *of eigenfunctions and associated functions of the problem* (1)-(3) *forms a basis in* $M^{p,\alpha}(0,1)$, $1 < p < \infty$ *after eliminate any function* $y_{i,n_0}(x)$ *it is necessary and sufficient that the corresponding function* $v_{i,n_0}(x)$ *of the biorthogonal system satisfy the condition* $v_{i,n_0}\left(\frac{1}{3}\right) \neq 0$.*If* $v_{i,n_0}\left(\frac{1}{3}\right) = 0$, *then after the eliminating function* $y_{i,n_0}(x)$ *from the system, obtaining system does not form basis in* $M^{p,\alpha}(0,1)$, *moreover in this case it is not complete and not minimal in this space.*

In (7) and (8) the parameter m which included in the problem (1)- (3), generally speaking is a complex number. But in some particular cases it is possible to refine the root subspaces of the operator L. So, if $m > 0$, then the operator L linearized of the problem (1)- (3) is a self-adjoint operator in $L_2 \oplus C$, and in this case all the eigenvalues are simple and for each eigenvalue there corresponds only one eigenvector. If $m < 0$, then the operator L is a J -self-adjoint operator in $L_2 \oplus C$, and in this case applying the results of [29,30], we obtain that all eigenvalues are real and simple, with the exception of, may be either one pair of complex conjugate simple eigenvalues or one non-simple real value. In the case of a complex value m the operator L has an infnite number of complex eigenvalues that are asymptotically simple and, consequently, the operator L can have a finite number of associated vectors. If there are associated vectors, they are determined up to a linear combination with the corresponding eigenvector. Therefore depending on the choice of the coefficients of the linear combination there are associated vectors satisfying the condition $v_{i,n_0}\left(\frac{1}{3}\right) \neq 0$, and there are also associated vectors not satisfying this condition.

Acknowledgement. This work was supported by the Azerbaijan Science Foundation-**Grant AEF-MCG-2023-1(43)-13/06/1-M-06.**

References

[1] A. N. Tikhonov and A. A. Samarskii. Equations of Mathematical Physics (Mosk. Gos. Univ., Moscow, 1999; Dover, New York, 2011).

[2] F.V.Atkinson, Discrete and Continuous Boundary Problems. Moscow, Mir, 1968.

[3] Collatz, L. Eigenvalue Problems. Moscow, Fizmatgiz, 1968, 504 p (in Russian)

[4] Gasymov T.B. Mammadova Sh.J. On convergence of spectral expansions for one discontinuous problem with spectral parameter in the boundary condition. Trans. of NAS of Azerb. 2006,vol. XXVI, no.4, p. 103-116.

[5] Gasymov T.B., Huseynli A.A. On the completeness of the eigen and associated functions of a discontinuous differential operator of the second order. Dokl. NAN of Azerb., LXIII, 2012, no.1, pp.3-7. (in Russian)

[6] Gasymov T.B., Huseynli A.A. The basis properties of eigenfunctions of a discontinuous differential operator with a spectral parameter in boundary condition. Proceed. of IMM of NAS of Azerb. vol. XXXV(XLIII), 2011, pp. 21-32.

[7] M. Shahriari. Inverse Sturm-Liouville Problem with Eigenparameter Dependent Boundary and Transmission Conditions, Azerb. J. Math., vol. 4, issue 2, July, 2014, pp. 16-30

[8] Bilalov B.T. Bases of Exponentials, Sines, and Cosines, Differ. Uravn. , 39.5 (2003), 619-623.

[9] Bilalov B.T. Some problems of approximation. Baku, Elm, 2016, 380 p. (in Russian)

[10] B.T.Bilalov, T.B.Gasymov, On bases for direct decomposition. Doklady Mathematics, 2016, v. 93, No 2, p. 183-185.

[11] B.T.Bilalov, T.B.Gasymov, On basicity a system of eigenfunctions of second order discontinuous differential operator. Ufa Mathematical Journal, 2017, v. 9, No 1, p.109-122.

[12] B.T.Bilalov, T.B.Gasymov , Maharramova G.V. On basicity of eigenfunctions of one discontinuous spectral problem in Morrey type spaces. The Aligarh Bulletin of Mathematics, 2016, vol.35, No.1-2, p.119-129.

[13] A. L.Mazuccato, Decomposition of Besov-Morrey spaces. In Harmonic Analysis at Mount Holyake, AMS series in Contemporary Mathematics 320 (2003), 279–294.

[14] Y.Chen, Regularity of the solution to the Dirichlet problem in Morrey space, J. Partial Diff. Equations 15 (2002), 37–46.

[15] P. G.Lemarie-Rieusset, Some remarks on the Navier-Stokes equations in , J. Math. Phys. 39(8) (1988), 4108–4118.

[16] Y.Giga, T.Miyakawa Navier-Stokes flow in with measures as initial vorticity and Morrey spaces, Comm. In Partial Differential Equations 14(5) (1989), 577–618.

[17] J.Peetre, On the theory of , J. Funct. Anal. 4 (1969), 71-87.

[18] Zorko C.T. Morrey space. Proc. Of the Amer. Math. Society, 1986, vol.98, No. 4, p. 586-592.

[19] B.T.Bilalov, A.A.Quliyeva, On basicity of exponential systems in Morrey type spaces, International Journal of Mathematics, Vol. 25, No. 6 (2014), pp.1-10.

[20] B.T.Bilalov ,T.B. Gasymov , A.A.Guliyeva, On solvability of Riemann boundary value problem in Morrey-Hardy classes. Turk. J. of Math. , 2016, v.40, No 5, p.1085-1101.

[21] T.B.Gasymov, A.A.Quliyeva, On basicity of the system of exponent with linear phase in Morrey-Lebesgue space, Advances in Analysis (AAN).

[22] Gasymov T.B., Maharramova G.V., Jabrailova A.N., Spectral properties of the problem of vibration of a loaded string in Morrey type spaces, Proc. of the IMM of NASA, Azerbaijan, v.44, no1.

[23] Gasymov T.B., Maharramova G.V., Mammadova N.G., Spectral properties of the problem of vibration of a loaded string in Lebesgue spaces, Transactions of NAS of Azerbaijan, Issue Mathematics, v.38, no.1.

[24] A.Q.Akhmedov, I.Q.Feyzullayev, On completeness of Eigenfunctions of the Spectral Problem, Caspian Journal of Applied Mathematics, Ecology and Economics V. 10, no 2, 2022, December, pp.43-54.

[25] Telman B. Gasymov, Alirza Q. Akhmedov, On basicity of eigenfunctions of a spectral problem in $L_p u \oplus C$ and L_p spaces, Baku State University Journal of Mathematics Computer Sciences 2024, v. 1 (1), p. 37-51

[26] Gasymov T.B. On necessary and sufficient conditions of basicity of some defective systems in Banach spaces. Trans. NAS Azerb., ser. phys.-tech. math. sci., math.mech., 2006, v.26, no.1, p. 65-70.

[27] Gasymov T.B., Garayev T.Z. On necessary and sufficient conditions for obtaining the bases of Banach spaces. Proc. of IMM of NAS.of Azerb.2007. vol XXVI(XXXIV). P. 93-98.

[28] Zigmund A. Trigonometric series. vol 2, Moscow, Mir, 1965, 537 p.

[29] Azizov T.Ya., Iokhvidov I.S. A criterion for the completeness and basicity of root vectors of a completely continuous J self-adjoint operator in the Pontryagin space $\prod_\kappa$. Math. iss., 1971, vol. 6, no. 1, pp. 158-161. (in Russian)

[30] Azizov T.Ya., Iokhvidov I.S. Linear operators in Hilbert spaces with a G metric. Uspekhi. mat. Science, 1971, vol. 26, no.4, pp. 43-92. (in Russian)

Telman B. Gasymov
Ministry of Science and Education of Azerbaijan, Institute of Mathematics and Mechanics, Baku, Azerbaijan
Baku State University, Baku, Azerbaijan
E-mail: telmankasumov@rambler.ru

Alirza Q. Akhmedov
Baku State University, Baku, Azerbaijan
E-mail: ehmedov-elirza@mail.ru

Reyhan J. Taghiyeva
Baku State University, Baku, Azerbaijan
E-mail: reyhanabasli2015@gmail.com

Received 05 November 2023
Accepted 11 December 2023